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Design of a Complete Height-Adjustable Workstation: An office solution Integrating the Frame, Tabletop and Accessories

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Abstract

This thesis investigates the design of heigh-adjustable desk as more integrated workstation for office workers. The study focuses on problems faced by the office workers such as cable clutter, limited storage, separate accessories and task related comfort while using the desk daily. The study follows a research-based design process by conducting literature review, market analysis, interviews and observations to find workstation related problems. These findings later are translated into design requirements and later used for concept development, prototyping and evaluation. The final concept includes a height-adjustable workstation with integrated tabletop, integration gap for accessories, storage, privacy, cable management and inclusion of sliding tray. The goal was to make these parts linked from the start instead of being separate add-ons. A working prototype was built and tested using several methods such as task-based usability, Rapid Office Strain assessment and System Usability Score feedback. The results showed low risk concern regarding ergonomic related to monitor and peripheral setup. Further research is required for long-term use and for production ready product. The study concludes with a product with system integration that supports office work better when cables, accessories, storage, privacy and ergonomic layout are considered together from the beginning.

Keywords – Office desk, Height-adjustable desk, Cable management, Accessory integration

SAMMANFATTNING

Denna studie undersöker utformningen av ett höj- och sänkbart skrivbord som en mer integrerad arbetsstation för kontorsarbetare. Studien fokuserar på problem som kontorsarbetare möter i sin dagliga användning av skrivbordet, såsom kabeltrassel, begränsad förvaring, separata tillbehör och komfort i samband med arbetsuppgifter. Studien följer en forskningsbaserad designprocess genom litteraturstudier, marknadsanalys, intervjuer och observationer för att identifiera problem kopplade till arbetsstationer. Dessa insikter omvandlades sedan till designkrav och användes därefter för konceptutveckling, prototypframtagning och utvärdering. Det slutliga konceptet består av en höj- och sänkbar arbetsstation med en integrerad bordsskiva, ett integrationsutrymme för tillbehör, förvaring, en integrerad skrivbordsskärm för visuell avskärmning, kabelhantering samt en utdragbar bricka. Målet var att dessa delar skulle vara sammankopplade från början, i stället för att fungera som separata tillägg. En fungerande prototyp byggdes och testades med hjälp av flera metoder, såsom uppgiftsbaserad användbarhetstestning, Rapid Office Strain Assessment och återkoppling genom System Usability Scale. Resultaten visade en låg ergonomisk risk kopplad till placeringen av skärm och kringutrustning. Vidare forskning krävs för att undersöka långvarig användning och vidareutveckling mot en produktionsklar produkt. Studien avslutas med ett produktförslag där systemintegration stödjer kontorsarbete på ett bättre sätt genom att kablar, tillbehör, förvaring, skrivbordsskärm och ergonomisk layout beaktas tillsammans redan från början.

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1. Introduction

1.1 Background

There is a significant increase in the modern office environment on computer-based work, resulting in prolonged sedentary behavior among office workers. Research shows that a substantial amount is spent by workers in the office by sitting, which has been associated with various health risks, including cardiometabolic issues and musculoskeletal discomfort (Phaswana et al., 2025). In health and ergonomics research sedentary behavior has therefore become a growing concern in workplace.

To address this issue, height-adjustable sit-stand desks have been introduced as an ergonomic invention that gives the user an alternative between sitting and standing positions. Some studies have even shown that these height-adjustable desks can guide employees to improved health, as they tend to promote physical activity during working hours and exhibit a sense of well-being within the office environment (Jindo et al., 2021). However, despite the various benefits, the sit-stand desks productivity remains limited due to various challenges in their long-term adoption, and design related barriers (Phaswana et al., 2025).

In addition to physical posture, the work conditions at the workplace can also affect an individual's perception of their work environment. Haapakangas et al. (2018) found that contentment with the physical environment, privacy and communication were strongly connected with self-assessed productivity and well-being in activity-based offices.

The organization of the workplace on the other hand plays a vital role in influencing user performance and experience. Research shows that an unorganized and cluttered work environment can act as physical and psychological stressors, causing increased cognitive load, low productivity and decreased tasks accuracy (McMains & Kastner, 2011). In addition to this, increased procrastination, job related tension and lower job satisfaction has been associated with higher levels of workspace clutter (Ferrari et al., 2018). Experimental studies

also indicate that individuals working in messy environments may exhibit lower accuracy in task performance compared to those in organized settings (Chae & Zhu, 2014).

Although there are some studies that suggest that disordered environments may support creative thinking (Vohs et al., 2013), the overall evidence indicates that excessive clutter can negatively impact focus, efficiency and well-being of the user. Despite this, many desk solutions together with accessories such as monitor stands, cable management systems and privacy screens look and treats as separate add-ons rather than an integrated component. This results in disconnected and inefficient workplaces that do not fully meet users' needs.

1.2 Problem Statement

Height-adjustable desks have become a staple solution to solve the negative effects of prolonged sitting in office environments. They offer flexibility by allowing users to switch between sitting and standing and they are often promoted as ergonomic improvement in the workplace environment. However, in practice the desks are not always used as they are intended as users do not use them regularly or stop using them over time due to issues related to comfort, usability and practicality (Phaswana et al., 2025).

The main drawback is that most existing desk solutions only focus on a single feature which is height-adjustability without considering the overall workspace as a system. In real office scenarios users rely on multiple accessories including monitor riser, cable management solutions, laptop stands and privacy screens. These accessories are usually added separately and are not designed to work together as a unified system. As a result, workspaces often become cluttered, visually inconsistent and less efficient to use. Previous studies on the design of desks have also investigated how lighting with the use of desk lamp, charging ports, storage and wire management can be integrated into one desk, which shows that functions can be treated as part of the furniture design rather than independent products (Wessam, 2020).

Office clutter is made up of various items found in the office that could either be work related or personal belonging of the person working there. The presence of such items can cause clutter and can distract the workers and make it harder for them to effectively work (Roster & Ferrari, 2020). Disorganized work environments can negatively affect focus, increase cognitive load and reduce productivity. At the same time, ergonomic benefits are limited because of poor workspace setup. This suggests that improving a single factor of the desk like height adjustability is not enough to support the needs in a modern office environment.

1.3 Aim

The aim of this thesis is to design an integrated workstation that is height-adjustable workstation for office workers, where the frame, tabletop and accessories are attached as one unit and considered as one coherent system. The project focuses on minimizing visual cluster, improving the placement of everyday accessories, supporting user task-related comfort, and creating visually consistent workstations suitable for modern office environments.

1.4 Height-Adjustable

Height-adjustability is one of the main features in this thesis as it allows office workers to change between sitting and standing positions during work. This is important because office workers spend a lot of working time sitting, which has made sedentary behavior a concern in office ergonomics and workplace health research (Jindo et., al 2021). The sit-stand feature is therefore used to support variation in postures for prolonged working periods. Ma et al. (2021) also found that sit-stand desk have shown to reduce sedentariness and improve some health and work-related outcomes, which include neck and shoulder, subjective health and work performance. However, the ability to adjust to height cannot be considered a complete solution. According to Phaswana et al. (2025), the continued usage of sit-stand desks can be affected by various range of issues such as discomfort, desk design, practicality, user motivation and workplace habits. Hall et al. (2019) also show that the use of sit-stand always

remains on how the user experiences the product based on their personal experience in that environment. This means that success of a height-adjustable workstation depends not only on lift function but also on how the whole setup of the desk supports everyday office tasks.

1.5 Accessories Integration

In this project, accessories integration means different elements around the desks are designed to work together as one system or connected to the desk to give visual feel of working together as one system. This includes accessories for cable management, monitor or laptop, storage system, privacy screen, access to power ports and other small accessories items. Keeping in mind that integration here doesn't mean incorporating everything into the desk itself rather making all the accessories look like an extension of the desktop. This is important because modern desktop is flat surface. Office workers use many different tools and objects during their workday, such as screens, laptop, headphones, phones, notes, documents and other personal items. When all these items are added separately the desk becomes visually messy and harder to use. Wessam (2020) shows that a work desk can integrated charging ports, storage, lighting and wire organization into one product, therefore supporting the idea that workstations can be designed as a part of furniture instead of being treated as separate add-ons. Integration of accessories is also related to privacy and workplace experience. Haapakangas et al. (2018) found that there was significant positive relation between environment satisfaction, privacy and communication with the ratings of productivity and well-being of workers in a work setting. This supports that privacy screens as well as other accessories should be integrated in office space where desks are shared. The integrated accessories in this project therefore are considered as a overall part of the desk concepts from the start. This helps the desk to give visual feel like one complete workstation rather than a normal sit-stand desk with several product added later.

1.6 Scope and delimitation

1.6.1 Include

This research focuses on the office workers using height-adjustable desks in daily computer-based work. The design work puts a special focus on the integration of tabletop, frame and accessories, with improvement in cable management, storage, privacy, comfort and visual coherence.

1.6.2 Exclude

The scope of work will not include electrical engineering analysis, motor system design, production costing, long-term health impacts or final industrial validation of the desk. The analysis will be limited to design input, low fidelity prototyping and ergonomic checks.
This thesis will not focus on development of height-adjustable leg system or structural mechanism. These parts will be considered existing solutions provided by the ROL Group. On the other hand, the project focuses on design related to desktop surface, integration of accessories and cable management solutions.
In this study, comfort will also be limited in scope as it will only include task-related workstation comfort. It includes things such as easy accessibility of often used objects, reduced unorganized placement of accessories, supporting sufficient workspace and reducing visual or cable clutter. The study does not evaluate thermal comfort, acoustic comfort, chair comfort or any long-term medical health outcomes.

1.7 Research Questions

1. What problems do office workers have with accessories, cables, storage and comfort in their current workstation setups?
2. How can integrated height- adjustable desk system be designed to reduce visual and cable clutter while improving task related user comfort?

1.8 Collaboration with ROL

ROL is a company in Sweden that design and develop several retail and office workspace solutions for other companies around the world. The project is done in collaboration with ROL Group. Therefore, the design focus is mainly on tabletop, accessories that can be developed as coherent workstation system around the existing height-adjustable frame.

1.9 Disposition

The thesis project is divided into different sections which are as follows:

1. Introduction – This chapter introduces about the background of the project, problem area, research questions, limitations and scope. It also includes that the project is done in collaboration with ROL Group.
2. Theoretical Framework – This Chapter caters to methods area which are relevant to the field of research. Some important pattern studies about workplace, sedentary lifestyle, workplace clutter. User-centred design and ergonomic evaluation tools
3. Methodology – This chapter shows how each method was used and conducted for the product development using the double diamond framework. It also describes the design approach, literature review, market and competitor analysis, interviews, observations, synthesis process, persona, concept generation and concept evaluation.
4. Results – In this chapter findings from the previous chapter are showed and they are combined into a synthesis table and translate into design requirements. The chapter also presents concept development and concept selection.
5. Final Concept – This chapter explains how the selected concept was developed further and how features of other concepts were included in the final design such as cable management, storage, accessories and privacy screen.
6. Prototype and Evaluation– This chapter explains how the working prototype was built and tested. How user feedback and ergonomic checks were done.

7. Discussion and Conclusions – This chapter shows the outcome of the thesis. It reflects on how the design answers the research question, what limitations were found and what could be improved in the future development
8. References – This shows all the sources and citation references used in the report
9. Appendix – This includes extra images that could not be included in the previous chapters

1.10 Application of Artificial Intelligence

Some artificial intelligence tools were used such as Chat GPT and Copilot in this thesis as supporting tools in the writing process. The tools were mainly used to structure ideas, improve the flow of language, sentences clarity, and organize research notes. The use of AI in this thesis was limited to structuring report content. It was not used to create the concepts or final design directions.

2. Theoretical Framework

2.1 Ergonomic in office work

Ergonomics is crucial to office layout design because office employees work for long periods at computer screen while performing routine tasks. In this case, ergonomics is not only about the physical position or posture of the body rather it can be defined as how the work environment also supports comfort and motion in completing routine tasks.

Jindo et al. (2021) describe ergonomics as system-based approach that includes physical, cognitive and organizational dimensions of work. This is important as the project does not only revolve around height-adjustable desk but also consider how the desk accessories, storage, cable management and user behaviour work together in one workspace. From this perspective, a desk should not only support posture but in addition to that also support the way people user their workspace.

Office workers are an important user group in ergonomics as they spend a considerable amount of their time working sitting. Jindo et al. (2021) explain sedentary behaviour is a health concern and office workers are a key target group whose sitting time shall be controlled in the workplace. This shows why workstation design can play a role in supporting healthier working habits. For this study, ergonomics is related with the support given by the workstation during the performance of daily repeated tasks in the office. This includes position of the monitor, keyboard, mouse, forearms and placement of frequently used objects. Hedge et al. (1999) showed that the geometry of the keyboard and mouse tray may affect the position of the wrists, seating position and discomfort of the upper body.

2.1.1 Workstation comfort

In this study, the workstation comfort refers to how easily an individual can perform their daily office tasks using the computer. This includes the placement of the monitor, keyboard, mouse, accessories and other things used during work. Emerson et al. (2021) explain that

computer workstations evaluation can help identify workstation problems that may contribute to musculoskeletal pain and can support recommendations for improving posture, comfort and safety. Therefore, the placement of the mouse, keyboard, monitor and accessories is seen as part of the workstation comfort in this study. WorkSafe Queensland (2020) recommends placing the keyboard and mouse directly in front of the individual, withing 10-15 centimetres from the edge of the desk surface. The monitor placement should be at arm's length away and regularly changing posture when using a sit-stand desk. Such recommendation highlights the importance of designing the workstations that will provide ease of reach to important items in both sitting and standing positions. In this study, comfort is therefore not to be understood as the broad feeling only. It is confined to only task-related aspects of comfort including reachability, ease of access of commonly used items, enough working space and reduced visual and cable clutter.

2.2 Height-adjustable desks and sedentary behaviour

Height-adjustable desks have become a common solution for reducing long periods of sitting in an office environment. These desks allow users to work while sitting and standing alternatively instead of sitting or standing at one position all day long. Phaswana et al. (2025) explain that height-adjustable sit-to-stand desks are often used to reduce sedentary behaviour among office workers. The same study also notes that these desks provide a practical workplace solution rather than interventions that require workers to exercise outside the office environment.

Jindo et al. (2021) also argue in support that office redesign for activity-based working and height-adjustable desks improves physical activity and sedentary behaviour. Their study showed that combine activity-based working and height adjustable desks in office environment had positive effects on some cardiometabolic risk factors, which implies that work environment design influence worker health.

However, height adjustable desks as a stand-alone cannot be viewed as a complete solution. Phaswana et al. (2025) mention that sit-stand desks interventions do not always lead to long term behaviour change. The long-term use can be affected by other factors like discomfort, practicality, workplace culture and design related issue, even if the users understand the possible benefits. This is crucial as it shows that a desk's success is not only dependent on its adjustability alone but also its ability to fit into a user's regular work routine.

2.3 Usability and adoption issues in sit-stand workstation

Even though sit-stand desks are designed to support healthier work behaviour, users may not use the desks as intended. According to Phaswana et al. (2025) participants dropped out of height adjustable work desk program due to various reasons such as discomfort, dislike, practicality issues and social perception at the workplace. This makes clear that the design of the workstation needs to be comfortable, practical and socially acceptable at the workplace.

A problem mentioned in the study was that some users felt the workstation was too bulky, large and difficult to incorporate into their existing work setup. Participants also reported problems while standing up, such as typing, reading, spreadsheets calculations and working with more than one monitor screen. This problem is relevant because it shows that a workstation must support real work activities, not just standing.

This also connects to the larger scope or idea of integration. Even though the desk may have a height adjustable option, its integration with monitor, laptop, cable storage and other accessories may not good enough. Thus, the design of an design of an adjustable height desk must consider the whole working station setup and not just the adjustable function.

2.4 Workplace organization and clutter

Organization of workplace in another important part of office design. A desk is much more than a tabletop surface for working as it is also a storage space for objects, documents,

gagets, wires and even personal items. When these elements are poorly organized the workspace can become cluttered and hard to use. According to Roster and Ferrari (2020), office clutter refers to artifacts whether work based or personal that leads to messy workspace. They explain that clutter is likely to distract employees and restrict movement, reducing a person's ability to work efficiently in workplace. This concept is highly relevant because many of the problems connected with desk setups are not only ergonomic but also related to organization and access. The same study shows, office clutter is an example of physical stressor in the workplace. The finding shows that clutter may have some associations with emotional exhaustion, decision making, productivity and overall, well-being. Therefore, clutter should not be treated only as visual problem as it can also affect people's emotion and how easily they manage their work tasks. Ferrari, Swanson, and Patel (2021) also studied clutter in remote working environments. According to their findings, the home office clutter was linked to job stress, emotional exhaustion, low job satisfaction, indecisiveness and procrastination. Thought the research focuses on the remote working environment rather than traditional office settings, the study is useful because it highlights how clutter in a work area can affect work behaviour and mental health.

In addition, the study highlights paper, office supplies and electronic items where commonly reported as sources of clutter in home office workplaces. This point is relevant for the present study because it represents exactly the type of objects that appear often around a desk and potentially require better organization and accessories for proper storage and management. For the present study, clutter is seen as both a practical and first-hand problem. Practically, it can make the desk harder to use and on the other hand, it can make the desk appear unorganized, unproductive and uncontrollable. This supports the need for a desk system that includes storage, cable management and accessories placement as a part of the overall design.

2.5 Office desk as part of a wider system

The office desk has been more than just a piece of furniture; it has acted as an important tool used to conduct business- and work-related tasks. According to Forino (2020), the office desks has always been linked to work habits, organizational systems and the development of office environments. In other words, this helped the desk to be viewed not only as a useful item but also a significant design element.

Forino (2020) also discusses cases wherein office furniture, interior design and architectural details were designed as part of a larger project. This is relevant for the present study because it supports the idea that the desk can be perceived as part of a system, especially when it interacts with accessories, storage, screen and workspace layout.

2.6 Ergonomic evaluation tools and standards

Ergonomic evaluation tools and standards have been used in this study to justify the evaluation of the workstation design. Since the study focuses on task-related comfort, reachability and posture support provided by the workplace, the evaluation aim is to check whether the design supports basic ergonomic principles during common office tasks.

2.6.1 Rapid Office Strain Assessment ROSA

This assessment is an ergonomic screening method developed for office and computer-based workstations. It is used to identify ergonomic risk factors related to the chair, monitor, keyboard, mouse, and general workstation setup (Sonne et al., 2012). ROSA is a picture-based checklist, which makes it suitable for observing or photographing users while they work at a desk. The final score ranges from 1 to 10, where a higher score indicates a higher ergonomic risk. A score of 5 or higher can be used as an action level, meaning that the workstation may need improvement (Sonne et al., 2012). In current study, ROSA is relevant because the project focuses on the design and evaluation of a height-adjustable office

workstation. The method is used to support the ergonomic evaluation of the functional prototype, especially for sitting posture, monitor position, keyboard and mouse placement, and the relationship between the chair and desk. However, ROSA does not fully evaluate the custom features of this project.

2.6.2 Task Usability Test

Task-based usability testing is a method of evaluating how the user interacts with the product while completing a specific task. This method also involves observing the user while they perform the given task, which makes it useful in a design project as some problems can only be seen when users perform the task in real time with the product. Usability can be understood by completion of the task by the user in a specific scenario (ISO, 2018). Bastien (2010) explains that behaviour is observed and documented in the user-based evaluation while the carries out the task.

2.6.3 SUS Method

The System Usability Scale is a validated short questionnaire with 10 questions that is used to measure how the users experience the product or system in terms of usability. It consist of ten statements developed by Brooke (1996) that users respond to on a 1-5 point scale, 1 being strongly disagree and 5 being strongly agree. SUS is helpful as it gives overall score in terms of usability without making the questionnaire too long and the final score is between the value of 0 and 100. It gives a measure of how usable the product or system appears to the user (Bangor et al., 2008).

2.7 User Centric Design

User centric design is important in this study since the aim is to design and cater for interaction between office workers and their desk in real situations. The design should not only be based on functions, but also on the needs, habits and problems of the users. ISO 9241-210:2019 explains that human cantered design focuses on the user, their needs and the

application of human factors and usability knowledge to make system more usable and useful. In this study user cantered design was applied through methods like interviews, observation and analysis of real workstations layouts. This helped with understanding problems related to cable management, storage, privacy, reach and comfort in a better way. The findings were later used to create a theme, design requirements and success criteria. This meant that design was not randomly chosen but was based one users' view and backed up literature. Human centred design supports early involvement of the user in the design process. Sternini et al. (2023) explains that using the end users at an early stage helps the designers to understand their impressions, needs and goals. According to Petrovic et al. (2023) human cantered design as a method helps align the product in the context of use. For this study, the workstation should be designed around the actual office including how office workers manage their tools placement, cable management and interaction with accessories during working hour

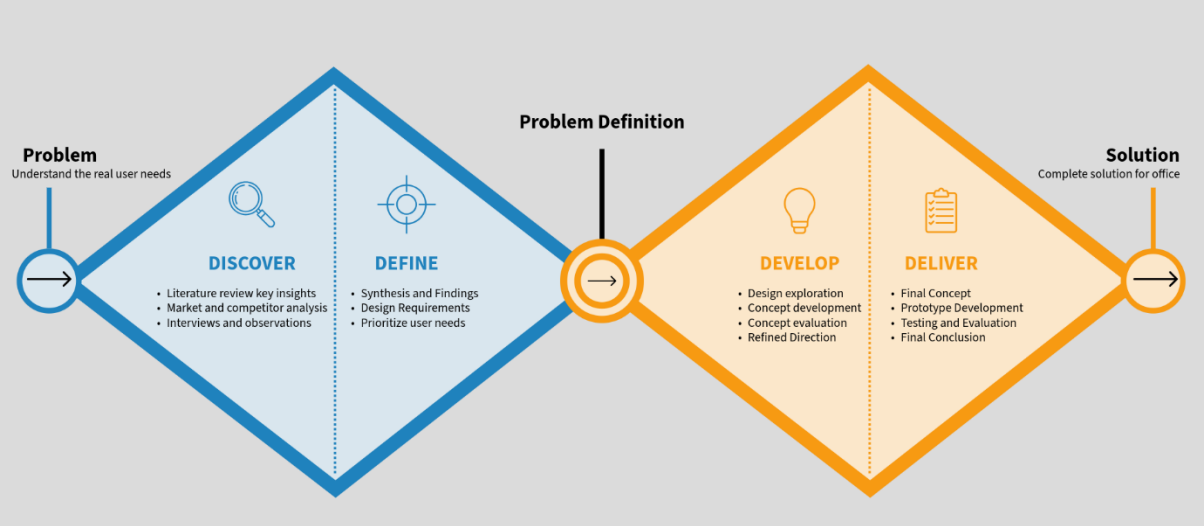
3. Methodology

This chapter explains the methods used in the current study and its purpose within the design development process for creating the final product. Since the current study is a design-oriented these methods were not just used to collect and contribute to information but also in translating the collected information into design requirements, concepts, prototypes and evaluation criteria. The study follows a research driven approach where first the problem area was explored through literature review, market analysis, interviews and observations. These results were grouped together according to themes and translated into design requirements. Thes design requirements were used as a guide to generate concepts and concept selection. Finally, a working prototype was planned and evaluated using usability evaluation and ergonomic check.

3.1 Research and design approach

Figure 1

Double Diamond Framework



Note. The 4 stages of the framework discover, define, develop, and deliver

The study follows structured sequence, starting from understanding the problem to developing and evaluating the design concepts. The first step focused on gathering and collecting information about the topic of office ergonomics, adjustable desks, work

environment organization and user behaviour. The second step focused on defining the main problem and translating them into design requirements. The third step included generating and selecting concepts, while the final stage focused on prototyping and evaluation.
Double Diamond framework was used as the organizing principle (Figure1 Double Diamond), The framework consists of four stages namely Discover, Define, Develop, and Deliver. The Discover and Develop stages involve exploration while the Define and Deliver stages helps to narrow down the project direction (Elmansy, 2021).
In this study, the Double Diamond framework was used to identify problems to designing and testing a concept. With the discovery phase data from literature review, market analysis, interviews and observation was collected. This helped to understand the current problems around the height-adjustable desks. In the Define phase, the collected information was analysed and converted into themes, needs, requirements, and criteria. This phase helped to create a clear design direction in which the design should be done. In develop stage, different design exploration was done, and 3 different concepts were explored via digital sketching and discussion with the company supervisor. The delivery stage, the selected concept was further refined, prototyped and used for testing and evaluation. The Double Diamond framework was useful as it kept the project structured while giving more space for exploration and iteration.

3.2 Discover

3.2.1 Literature Review

A literary review helps to give a summary of what is known, theories used, methods and gaps in the selected area of study (McCombes, 2023). In design research, this is important since literature is a source of information that influences the design process. According to Säfsten et al. (2020), a literature can be supported by the following process, identifying suitable keywords, specifying selection criteria, selecting suitable database, and creating a search strategy. In the current study, keywords were selected based on focus area of the

thesis. These included “height-adjustable desk”, “office ergonomics”, “workstation comfort”, “cable management”, “desk accessories", and “integrated workstation design”. Academic database and search engines such as Google Scholar, Primo, among others were used to find relevant articles, reports, books, and standard guidelines. In addition to this searches, backward snowballing technique was used. Snowballing is a technique in which one looks at the references lists of already selected sources to find other relevant literature (Wohlin, 2014). This helped to find more materials about ergonomics and workstation layout.

In the initial stage of the current study, a literature review was carried out to understand the main topics connected to the current study. Topics focused on office ergonomics, height adjustable desks, workspace organization, activity-based workstations, and workstation layout design were picked.

The purpose of the literature review was not simply to collect information about the topics background rather it was aimed at identifying and discovering findings that could later support the design decisions and requirements. For example, studies on height adjustable desks helped in understanding the importance of sit-stand desks, whereas studies on office clutter and workstation design confirmed the necessity of storage space, cable management and accessories organization.

The selected sources included peer-reviewed research articles which were mainly used for ergonomics guidelines, clutter and workstation comfort. Design theses were occasionally used mainly as supporting references for workstation design examples and design direction. The literature review’s result were paired with the findings of market analysis, interviews and observations to create the synthesis table, design requirements, and success criteria.

The main literature reviewed were:

(1) Office ergonomics

(2) height-adjustable desks

(3) sit-stand workstations

(4) workspace clutter and organization

(5) activity-based and flexible workstations

(6) keyboard mouse, forearm support and reach zones

(7) ergonomic evaluation tools such as ROSA

3.2.2 Market Research and Competitor Analysis

Market research is defined as a method used to collect information regarding specific market, product category, users and competitors. It helps the researchers to understand the existing product in the market by giving the insight of what if available at present, what customer needs, the gap in the market, and possible opportunities for developing new products. As explained by Bhat (2024), “market research can be considered as organized process of collecting data for the purpose of making better decisions and understanding how the target audience may respond to a product or service.

Market and competitor analysis were used to examine the current height-adjustable desk and their related workstation solutions. The purpose was to understand how these products currently solve the associated problems with cable management, storage issues, and accessories integration. This was done by analysing different desks and workstations based on the above-mentioned problems putting them as criteria and checking which features are missing and which features were added from the start. If there were features such as storage, cable management in some existing solutions it was checked and noted down how they were integrated. This was also important because the findings from this method were later used together with the literature review, interviews, observations to create a synthesis table and guide the concept development phase.

3.2.3 Interviews

Interviews were done to understand how office workers use their desks and what problem they face in their daily work setup. Interviews can be used during design process because they help the designer understand people's perspective, values, needs, and motives connected to product or situation (van Boeijen et al., 2020). The interviews helped to collect user perspectives on matters related to comfort, organization, cable management, privacy, storage and sit-stand working habits.

The semi-structure format was used during the interviews of the participants. This allowed a prepared set of questions but also provided more space for elaboration by the participants to explain their own experiences in more detail. This was important to obtain information on real user behaviour and not collect short answers.

The interview questions focused on:

(1) Current desk setup

(2) use of desk accessories

(3) cable management problems

(4) storage needs

(5) comfort during work

(6) sit-stand working habits

(7) privacy needs

(8) problems with clutter

(9) desired improvements in a future desk system

The data obtained during the interview were later compared with the observation's findings. This helped identify reoccurring problems and user needs, which were later translated into design requirements.

The total number of participants were 8 office workers, and the interview was conducted in

the offices of ROL Group company and Jönköping University both situated at Jönköping, Sweden. The duration of the interview was 20 mins approximately for each participant. The questions were prepared from before and the answers were recorder through pen and paper. Semi-structured questions were used during the interview.

3.2.4 Observations

User observation is a methodology used by designers to help recognize individuals, contexts and various basic interactions in everyday situations. Direct observation can reveal unwanted and unanticipated usage situations, this developing a deeper knowledge of what constitutes a successful product. Designers can enhance his design by analysing the interaction between the individual and the product. It also gives rich visuals whenever the designer communicates their ideas on designs with different people (Boeijen, 2020). Observation was taken place at the same places where the interviews were held. Observations were done simultaneously while conducting the interview to understand how user organize their desks and interact with their desk. This method was useful as some problems were easier to see than by interviewing someone verbally. For example, object placement, lack of accessibility to frequently used items, cable clutter and poor accessory placement can often be noticed through visual observation.

During the observation, attention was given to:

(1) Monitor and laptop placement

(2) visible cable routing

(3) objects placed on the tabletop

(4) storage habits

(5) personal items

(6) accessories used during work

(7) reach distance to frequent used items

(8) privacy needs

(9) overall organization of the workstation

Consent of each participant was taken before starting the interviews and observation. Photographs of their desk setups were taken. These photos were used for analysis and presentation within the study. The findings and photographic evidence helped support the interview data and gave a clear picture of real desk use.

3.3 Define

3.3.1 Data analysis and synthesis

After finishing the literature review, market analysis, interviews and observations, the results were grouped into specific themes. This helped in moving from raw information to clearer design direction. Results were grouped based on specific problems and needs. Common patterns and findings were grouped together to identify trends. The main theme that came out of the research was:

(1) Cable routing and power access

(2) Accessories and installation

(3) storage and personal items

(4) Comfort, reach and posture

(5) Visual calm and coherent appearance

These themes were then used in the synthesis table. The table included research, user needs and design requirements and potential design criteria. This was crucial as this showed the design requirements are a result of research findings rather than being chosen randomly. For example, findings about visual cable and clutter resulted in requirement of integrating cable management and storage. Finding related to posture and comfort were translated into requirements about monitor, keyboard and mouse placements and reach zones.

3.3.2 Design requirements and success criteria

Design requirements were created and picked up from the synthesis table. There requirements were used in the design exploration and concept generation phase to guide the concept development and to check whether the design direction responded to the current study needs

3.3.3 Persona

Developing persona helps designer understanding customers' requirements, behaviour and objectives. This helps in recognizing that every individual requirement and desire is different. Persona also helps in achieving positive consumer satisfaction for the target audience. The process of creating persona for regular customers enables the understanding of the results obtained regarding the kind of people the designer wants to design for (Siang, 2025).

3.4 Develop

3.4.1 Sketching

This method can be referred to as a unique tool that creative professionals use to provide, discover, improve and express their ideas. The drawings done by sketching are quick and easy to modify. Sketching avoids formal rules unlike verbal and written communication, which helps the user to convey their ideas easily and avoid misunderstandings. It is possible to modify and analyse the chosen concepts using another drawing to ensure that the selected idea is the most suitable approach for addressing the given problem (Rojas, 2023).

In this study, sketching was done initially after getting the desired design requirements from the synthesis table and persona. It was used to explore different forms and design that could work together. The sketching was mainly done digitally to be more effective. In the later stages, sketching was used to develop concepts of workstation to see which one fits the best and later the concept selected was also once again refined by sketching to get a final look.

3.4.2 Concept Generation

Concept generation is a part of the product development through which different direction of solutions is explored before picking the final concept (Ulrich & Eppinger, 2016). Concept generation was the next stage after the key findings and requirements had been defined. The aim was to explore different approaches to designing an integrated height-adjustable desk. Different concepts were explored using sketches, digital sketches and feature combinations. Each concept focused on a unique way of solving the main problems identified in the synthesis table. The concept generated presented different ideas regarding cable management, mounting accessories, modular panels, storage, monitor riser, laptop mount, sliding tray and incorporation of the privacy screen.

3.4.3 Concept Evaluation

Concept selection was done in collaboration with the company ROL Group in case of this project, as they serve as the client for the purpose of this thesis. This means that people of ROL Group provided feedback on the presented concepts and ultimately decided which design features will be implemented in the final concept. Concept selection is a stage in product development in which different solutions are compared, and the most suitable direction is picked for the further development (Ulrich & Eppinger, 2016).

3.5 Deliver

3.5.1 Working Prototype

To test the final concept a working prototype was made of almost the same proportion as the final concept. To allow the user to physically test prototyping is useful before the product is fabricated into final product (Ulrich & Eppinger, 2016). In this study, working prototype was used to evaluate the usability of the workstation.

3.5.2 Task-based usability testing

Task-based usability testing was done to get feedback and observe how the users interacted with the desk while performing a given task. It is helpful as this allows the designer to observe user while using the product rather than only taking their opinion about the product (Rubin & Chisnell, 2008).

3.5.3 Rapid Office Strain Assessment

A partial Rapid Office Strain assessment was used to check the level of risk of ergonomic of the workstation. ROSA method is used to quickly check ergonomic risk factor in office workstations setups especially for computer-based work (Sonne et al., 2012). In current study, the test was partial as the chair was not included because the chair was not part of the designed product. The test focused on monitor, keyboard and mouse related reach and posture problems.

3.5.4 System Usability Scale

System Usability Scale questionnaire was used to collect responses on the usability of the working prototype. SUS is developed to provide quick overall score for usability after the user interacted with the system or product (Brooke, 1996). In this study, the SUS score gave early feedback to understand users' perception on usability of the desk system.

4. Results

This chapter shows the findings from the methodologies used in the Methods Chapter. The results are based on literature reviews, market and competitor analysis, interviews, observations. The information from these findings were later grouped together and combined in the synthesis table, which translates into design requirements for the concept generation phase. The purpose of this chapter is to not only show the findings from each method, but also to show how the research helped in defining the main problems connected with cable management, storage, accessory integration, task related comfort, reachability, and visual clutter.

4.1 Literature review

The literature review was done to understand the main topics connected to the current study, such as ergonomics, height-adjustable desks, workstation comfort, sit-stand adoption, clutter, storage issue, cable management and integrated workstation design. The purpose was to understand the design directions which can be later translated into design requirements.
The literature showed that workstation design is not only about sit-stand feature. It also depends on how the desk support daily office tasks for the user. The placement of the monitor, keyboard, mouse, accessories and frequently used tools influences the comfort factor. Emerson et al. (2021) explains that workstation helps in identifying problems that contribute to musculoskeletal pain and support for better posture, safety and comfort. WorkSafe Queensland (2020) also recommends that keyboard and mouse must be placed in front of the individual, while monitor placement must be at arm's length from the body. Studies related to posture of the forearm were also included to understand the placement of the keyboard and mouse and how it influences workstation comfort in relation to computer-based office work (Cook et al., 2004).

The findings from the existing literature also showed that height-adjustable desks could help in sedentary behaviour and can bring changes in postures. According to Jindo et al. (2021) office workers with height-adjustable desks have potential risks of having sedentary behaviour. Similarly, Ma et al. (2021) also found that interventions including sit-stand desks could decrease in overall sedentary behaviour and improve health aspects and performance at work. At the same time, it is also clear from the literature that simply having the sit-stand feature is not enough. According to Phaswana et al. (2025) and Hall et al. (2019), the sit-stand desks use depends on office culture, comfort, user behaviour and how well the desks fit into daily work tasks.

Another finding was connected to clutter and organization. Roster and Ferrari (2020) describe office clutter as work-related or personal objects that leads to disorganized workplace can distract user or reduce their ability to work effectively. This study is relevant to the project since accessories and personal items are also a part of the workstation, and they should also be considered from the initial design process. Dao & Ferrari (2020) also found that clutter inside the office affect the well-being and job satisfaction, which showed the design direction for better overall organisation in the workstation design.

The studies also stress the need for privacy and visual coherence. Haapakangas et al. (2018) found that overall wellbeing was linked with the quality of physical setting, privacy and connection within activity-based offices. Wessam (2020) also shows a desk where charging ports, storage, lighting and cable management are integrated into one product. This relates to the idea that all functions could be designed as a part of the desk rather than added later as separate products. There were other studies related to design that supported the desk as a flexible system. Henriksson and Nordström (2020) show desk and workstation design can respond to the needs of changing workplaces. Rogberg (2021) explains the importance of workspace layout, user needs and practical use in office workstation design. These studies

show that desk should not designed as only be a flat surface rather it should be a connected workstation. Overall, 14 relevant sources including peer-reviewed journal articles, master's theses and ergonomic guidelines were reviewed. The literature review showed that design should focus on a complete solution of the workstation. Table 1 below shows the main findings and design implications when the sources collected were grouped as themes. These finding were later combined with other methods to create a synthesis table and design requirements.

Table 1

Literature Findings

S. No	Literature theme	Main finding	Design implication
1.	Ergonomics and workstation comfort	Workstation layout affects posture, reach and comfort during computer-based work.	Monitor, keyboard, mouse and frequently used accessories should be placed within comfortable reach
2.	Height-adjustable desks	Sit-stand desks can reduce sitting time and support posture variation.	Height-adjustability should be included, but it should be treated as one part of the whole workstation.
3.	Adoption and usability	Users may stop using sit-stand desks if they feel uncomfortable, impractical or difficult to use.	The workstation should support real office tasks such as typing, screen use, cable access, storage and accessory use.
4.	Clutter and organization	Clutter can distract users and reduce the ability to work effectively.	Cable management, storage and object placement should be integrated into the desk from the beginning.
5.	Privacy and work environment	Privacy and the physical environment can affect productivity and well-being.	The privacy screen should be part of the workstation system and not only a separate add-on.
6.	Integrated desk functions	Desk-related functions can be integrated into the furniture itself.	Accessories, power access, cable routing and storage should follow one coherent design language.

4.2 Market and competitor analysis

The market and competitor analysis was done to understand how existing desk system are being used and what all accessories can be integrated in them. The cable management and

storage were also analysed and compared to get a better understanding if that desk system has this feature or not. The criteria of the table were including accessory interface, cable management, integrated storage and general design notes from each desk.

To maintain confidentiality of some competitors of ROL Group the names were changed to System A, System B and so on. Figure 2 below shows some of the existing workstations in the market.

Figure 2

Existing desks on the market



Note. Images collected from publicly available product and market websites and used only for visual analysis in this thesis.

Table 2

Findings from Market and Competitor analysis

System (Anonymous)	Accessory interface (no clamps?)	Cable management integrated?	Storage integrated?	Notes
System A	O	O	O	Height-adjustable desk with accessory ecosystem mainly through separate add-ons. (market/competitor)
System B	—	✗	✗	Premium desk, accessory/storage ecosystem not clearly evident in reviewed material. (competitor)
System C	✗	O	✗	Office desk system with privacy/screen, cable solutions exist but are not strongly integrated (market/competitor)
System D	—	O	✗	Basic sit-stand desk with limited cable organization, no clear integrated storage. (competitor)
System E	O	O	O	Modular through optional accessories rather than built-in integration. (competitor)
Reference 1	✓	✓	✗	Strong integration of power, cable management, and accessories, not a typical office-desk (Reference)
Reference 2	✓	O	O	Strong modular attachment logic and organization, useful as a reference for add-on integration. (Reference)

Note. ✓ = clearly present, O = partly or optionally present, X = not evident

The table 2 shows analysis that existing desks in the market offered cable management, but it was either optional to upgrade or partial integration where the solution is often hiding the cables under the desk. This can make the desk look cleaner from the front, but the cables are not easy to access or adjust. In some cases, the desk had a rear tray component under the desk which acted as a cable management system, but it was not fully connected to the rest of the desk or was an add-on to upgrade.

Another finding was accessories were often treated as an add-on. Some desks allowed accessories to be attached through rails and clamps, but connecting accessories to the desk with these didn't feel like a natural part of the desk. This creates a gap between function and integration Individually the accessories may work, but they can still look visually separate from the desk system.

Storage as a feature was also limited in many reviewed desks systems. Some workstations did

include storage options, but were often optional, separate or not clearly integrated into the main desk structure. For example, if there was a storage option available for upgrade it was a stand-alone cabinet which was not a part of the desk itself.

The market analysis revealed that premium desks often have strong visuals, however adding further accessories may cause loss in visual unity. This is significant point for the current research since the purpose of this study is not just enhancement where bunch of accessories are added but to consider the features as a part of the desk from the beginning.

4.3 Interviews and Observation

Interviews and observations were done to understand the user's perspective in real situations. The observations were crucial as they showed some problems that users may not aways mention directly in interviews.

The interview were conducted in the company office of ROL Group and in Jönköping University school both situated in Jönköping Sweden. In total of 8 participants were interviewed and observed. The images of their desk setups were taken after getting the consent of participants.

The interview was semi-structured, and the questions asked were as follows:

1) Tell me about your typical workday at your desk
2) What task do you do most at your desk
3) How often do you switch b/w sitting and standing?
4) What items are always on your desk? What comes and goes?
5) What devices do you use daily?
6) Which desk accessories do you use right now? Why did you add them?
7) Which accessory is most important, and which will be nicer to have?
8) What do you not like about your accessories?

9) Show me how the power and cables are managed? What parts feel messy and annoying?
10) Where does your personal item go (bag, headphones, bottle, documents, phone)
11) What ends up on the desk even if you don't want it there? Why?
12) Do you ever feel discomfort while working on your desk? Where? Cause? Fix?
13) How often does your workplace change?
14) If complete desk solution came with built in accessories and cable management, what would you expect?
15) If you could fix one thing about your desk setup, what would it be?
16) Rank these in order of priority cable mess, accessories, storage, comfort, and aesthetics

Figure 3

Desk setups of office workers at ROL Group



From the above figure 3, the observations showed that cable management was a reoccurring pain point and was also listed as the top priority for the participants when ranking their priorities during the interview. Cables were often visible under the desk, behind the

monitors, gathered in an unorganized way and in some cases hanging below the tabletop. This gave the workstation a messy look and made it harder to adjust the setup. Storage and organization of the objects on the tabletop was also a repeated issue where users kept their personal, daily in use items on the tabletop, such as pens, mugs, bottles, notebooks, phones, headphones and other documents. Some participants kept their personal items like bags on the floor or under the desk because they did not have a storage space, causing visual clutter to the workspace. Another crucial thing participant mentioned and relied heavily on separate accessories and add-ons. For example, monitor arms, laptop stands, wrist supports, desk lamps, privacy screen and cable clips. Even though these were solving their individual problems, but they were not connected to desk visually or functionally. Observation showed that even with the height-adjustable feature comfort issues related to reachability and poor placement of the items affected the posture. This finding supported the need to consider the workstation layout as a whole rather than only focusing on sit-stand feature.

Key findings from the interviews and observations were

(1) Cable management is a repeated pain point

(2) Better integrated storage is needed

(3) Placement issues with objects

(4) Users rely on separate accessories and add-ons

(5) Desk surface becomes visually clutter over time

These findings confirm the main problem was not only the desk itself, but all these listed things combined.

4.4 Persona

A persona was created based on the previous methods to summarize the main user group. The persona helped to make the user needs easier to understand during concept generation.

Figure 4 below shows the top priorities of the primary user group with their goals, office behaviour and habits. The persona's main goal was to work comfortably for long period while keeping the desk clean and organized. Users keep items such as phones, mugs, bottles and documents on the desk. They may keep their personal belongings such as bags on the floor, use one or more monitor with a laptop and added accessories later depending on their problem. The desk surface was used as a temporary storage and device zone.

The top priorities for the user were:

(1) Cable management

(2) Comfort

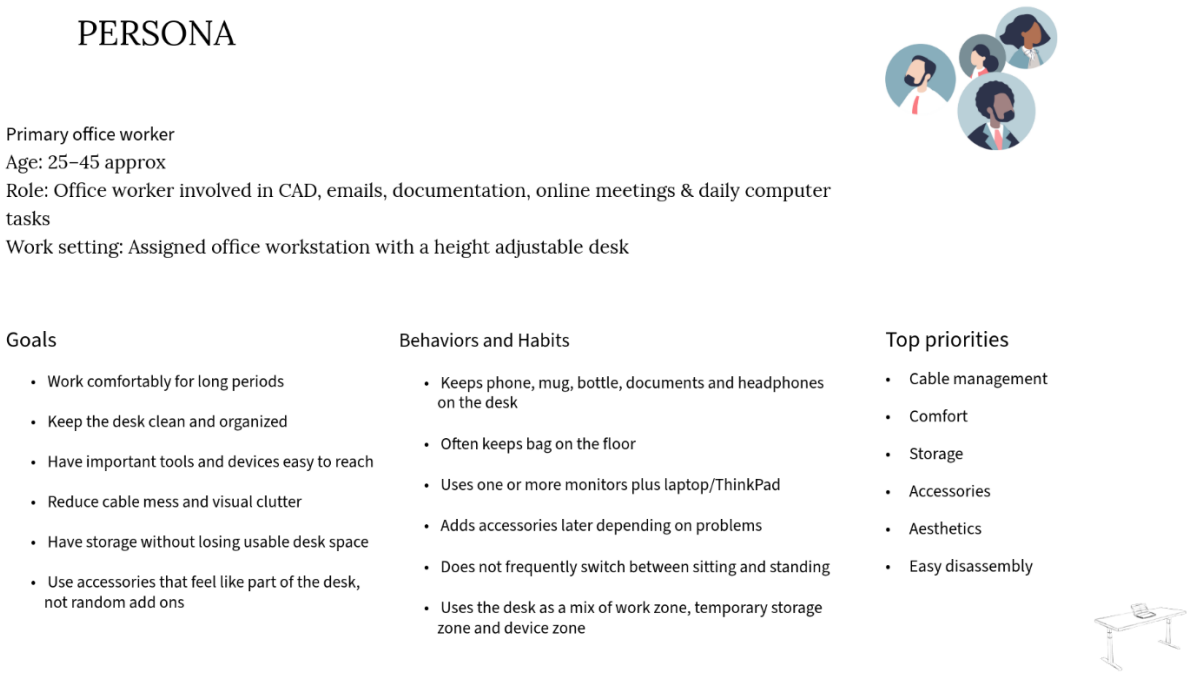
(3) Storage

(4) Accessories

(5) Aesthetics

Figure 4

Persona



4.5 Data analysis and synthesis

Table 3

Synthesis table

Theme	Literature Review	Market / Competitor Analysis	Interviews & Observation	Design Implication / Requirement	Priority
Cable management	Cable clutter reduces visual order and can negatively affect usability and perceived workspace quality. Easy organization improves desk experience.	Almost all desks try to solve cables through trays, rear compartments, under-desk channels, or add-on systems. Cable management is common, but often hidden at the back and not always easy to access.	Users repeatedly mentioned messy cables, chargers, and device wires. In observations, cables were often visible, hanging, or gathered in an unorganized way.	The desk should include a clean and accessible cable management solution that reduces visible clutter without making access difficult.	Must have
Visual clutter / clean workspace	A visually organized workspace supports focus, comfort, and better overall user experience.	Many premium desks present a minimal look, but extra accessories often still appear as separate add-ons.	Users wanted a cleaner setup and less mess on the tabletop. Photos showed crowded surfaces with daily-use items competing for space.	The design should help create a visually calm and organized workspace. Accessories should support order, not add more noise.	Must have
Integration of accessories	Integrated systems often provide a more coherent user experience than disconnected add-ons.	Most desks allow external accessories, but fewer create a truly unified system. Accessories are often optional and visually separate.	Users need monitor support, laptop placement, cable help, storage, and small utility functions. These needs exist together, not separately.	Accessories should feel seamless and coherent with the desk system, not like random attachments. Integration here should be visual and functional, not necessarily fully built into the desk.	Must have
Flexibility / modularity	Different users have different workflows, so flexible systems are valuable when they remain easy to understand and use.	Some desks support modular add-ons, rails, pegboards, or optional accessories. However, too many separate parts can become complicated.	Users had different priorities and desk habits. Not everyone needed the same setup, but many wanted the freedom to adapt their space.	The desk should support some level of modularity or customization so users can adapt it to their workflow.	Should have
Accessibility and ease of use	Features that improve organization must also be easy to reach and simple to use, otherwise users avoid them.	Rear cable solutions are common, but not always convenient during daily use. Some solutions look clean but may be less accessible.	Users prefer solutions that are practical in daily work. If storage or cable access is difficult, people stop using it properly.	Important functions should be easy to reach, easy to understand, and convenient in everyday use.	Must have
Tabletop utility	The desk surface is not only a flat plane; it can support workflow, organization, and ergonomic interaction.	Competitor desks mostly focus on surface area, cable tray, and accessory add-ons. Fewer explore the tabletop as an active functional zone.	Users keep important everyday items on the desk because they need quick access. This often increases clutter.	The tabletop should offer better utility, for example through zones, support features, or integrated organization elements.	Should have
Storage / organization of small items	Workspace organization improves when frequently used items have a defined place.	Some desks include drawers, shelves, pegboards or under desk storage but storage is not always well integrated.	Observations showed pens, notebooks, chargers, headphones, and personal items spread across the desk. Users need quick access without losing order.	The design should support light, everyday organization for small work tools and personal items.	Should have
Ergonomic support	Desk solutions should support comfortable reach, posture, and healthy use during work. Ergonomics is affected not only by desk height but also by placement of tools and accessories.	Most competitors focus on sit-stand adjustment, but fewer address ergonomic positioning of accessories as part of an integrated system.	Users use monitors, laptops, keyboards, and desk items in different ways. Poor placement can affect comfort and posture.	The concept should support ergonomic positioning of accessories and work tools, especially within normal reach and comfortable use zones.	Must have
Aesthetic coherence	Product experience is influenced by both function and appearance. A coherent aesthetic can improve perceived quality and acceptance.	Premium brands often use a strong visual language, but once many accessories are added, the setup can lose unity.	Users may not always say "aesthetic coherence" directly, but they respond positively to desks that feel clean, balanced, and intentional.	The final concept should have a consistent design language so all elements feel like one family.	Could have / Should have

The findings from the previous methods such as literature review, market analysis, interviews and observations were combined into a synthesis table. The purpose of the synthesis table was to give design requirements from the research findings. This made it easier to trace the design process since each requirement had its own origin which could be traced back to a research input.

From the above Table 3 Synthesis Table, cable management was identified as a must-have requirement. Literature and observations showed cable clutter can reduce visual order and negatively affect the workspace environment. The market analysis showed cable management is always at the back of the desk in the existing desks and often hidden. Therefore, the desk requires proper cable management to avoid mess and remain accessible.

Visual clutter and clean workspace were also identified as a must-have requirement. Users want a clean desk surface with less mess on the tabletop. Observations photos showed messy and unorganized surfaces where daily items were left on the desk. Therefore, the design should create a visual calm and organized workspace. Accessories should not add more visual noise rather support the desk. In this study, accessories not necessarily means that it should be in built within the desk instead it means that the accessories should feel connected to the desk visually and functionally.

Flexibility and modularity were also identified as a should-have requirement. The users have different desk habits, so the desk should support some level of adaptation. However, research also shows that too many isolated parts can be complicated. Hence, modularity should be there in a controlled and understandable way.

Both accessibility and ease of use were identified as must-have feature requirements. If a feature that helps in organization is difficult to reach or understand it is not a useful feature. The desk needs to make job easier by making important functions easy to reach, especially when dealing with storage and accessories.

Another should-have requirement was tabletop utility. The desk is not just a flat surface as it can support workflow, organization, and ergonomics. This means that tabletop include zones, support features, or organization components without reducing the amount of usable workspace.

Storage and items organization were also identified as should-have requirements. Observations showed that for quick access users often leave their work tools or personal items on the desk. This gave the direction of having small storage that is easy to use.

Ergonomic support was identified as a must-have requirement. Findings from every method showed that users work with items such as monitors, laptop and keyboards in different ways. Poor placement of these items affects posture and comfort for reach. This gave the design direction that support comfortable reach zones for better ergonomics positioning.

Aesthetic coherence was identified as a should-have or could-have requirement. Users described that desks should feel clean, simple and balanced. The final concept should have this design direction of consistent design language so that every element feels like a small part of a big part.

4.6 Summary of design requirements

With the synthesis table came design requirements that were used for the next phase of the study. The most important requirements were:

(1) The desk must include an accessible cable management

(2) The desk must reduce visible clutter on and around the workstation

(3) Accessories must feel integrated with the desk system

(4) Important functions must be easy to reach, understand, and use

(5) The desk must support ergonomic placement of monitors, keyboard, mouse and other work tools

(6) The tabletop should provide better utility through zones, storage or integrated organization

(7) The tabletop should prove storage for small work tools and personal items

(8) The design should allow some flexibility or modularity so users can adapt it to their workflow

(9) The final design should have a consistent visual language and feel like one coherent system

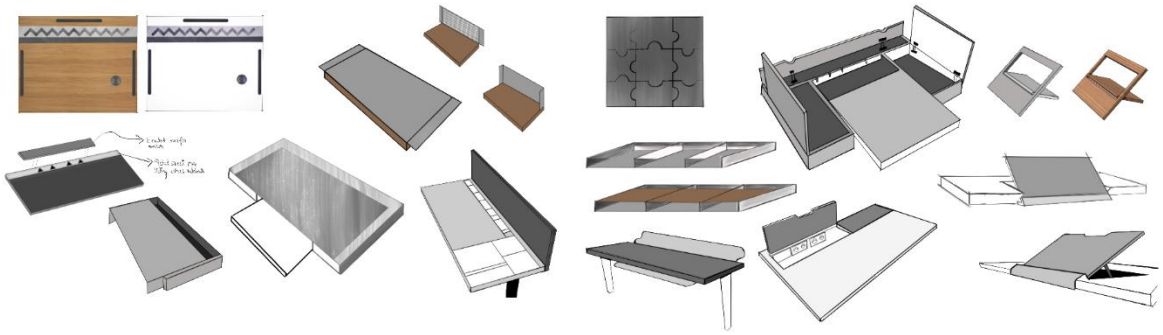
The concept development phase was based built on the above list of requirements as they served as a guideline in designing solutions and helped in determining whether the concepts solve the actual problems found through the research.

4.7 Concept Development

4.7.1 Design exploration

Figure 5

Design exploration



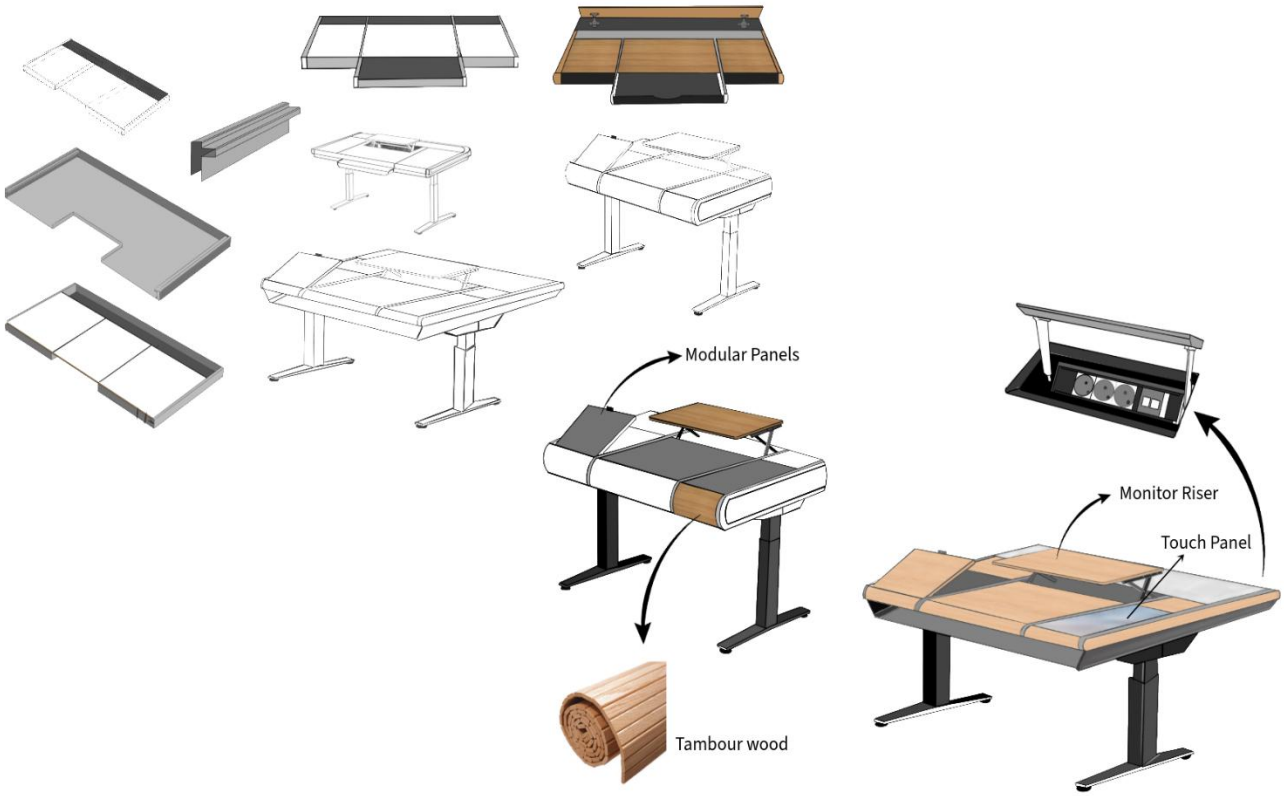
The figure 5 above show the design exploration, which was the first step towards sketching the concepts where free hand sketches were done on pen and paper or digitally on iPad. The idea was to sketch roughly to generate some solutions before going straight to sketching a proper concept.

Before jumping straight into generating concepts some rough sketching was done digitally and on pen and paper. For digital sketching software Procreate and device iPad was used to quickly draw some ideas and modify them for improvements. The design exploration phase helped in creating multiple concepts.

4.7.2 Concept 1

Figure 6

Concept 1



The above figures 6 show the sketch development of concept 1. This was the first concept which explored the modularity of the heigh-adjustable desk. The theme of the desk was to divide the tabletop area into functional panel instead of treating it as one fixed surface. These panels can be lifted, removed and opened to hidden storage. The direction of clean

workstation with functional elements was used in this concept where all the elements are integrated into the desk rather than added as separate external products.

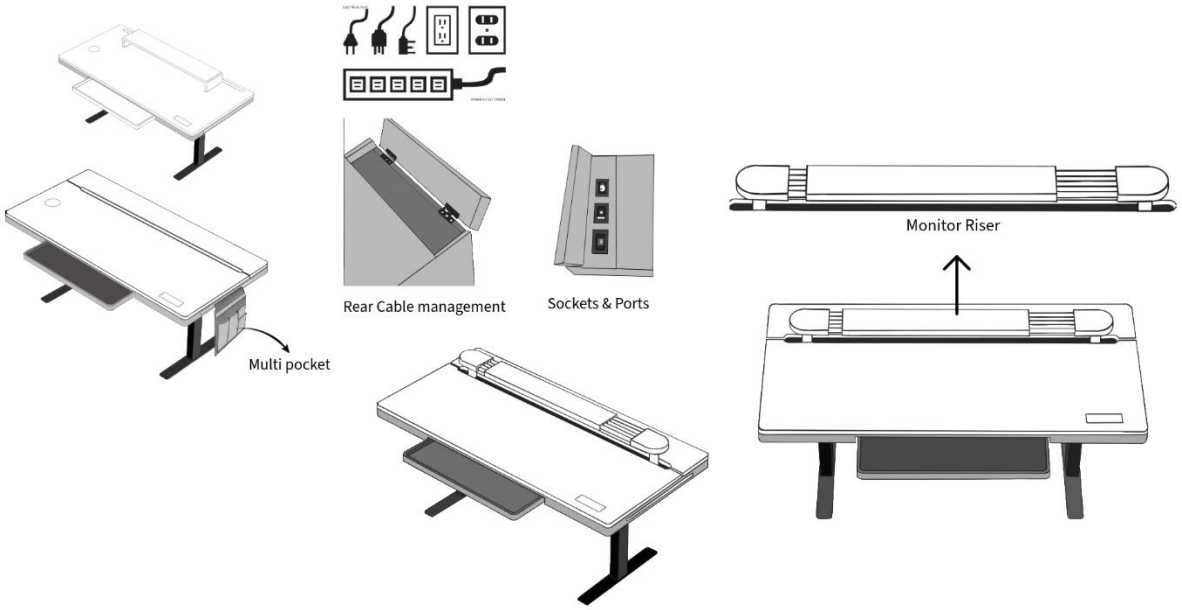
The wooden panels are on top of a metal frame which gave the tabletop the underneath support and the metal frame also acted as a cavity for storage. A tambour wood intended detail was considered for the curved surface front, as it follows a rounded form while still giving the desk a softer and more furniture appearance. A key feature of this concept was the monitor riser with a simple mechanism of lifting and down. It allows the user to raise the screen of the monitor up when needed for a better viewing height and lower when not in use to give the surface a clean look. The concept also includes a touch panel integrated into the tabletop to control the desk. The placement of the touch panel was near the proximity area of the user. The rear end of the desk was designed for the cable management zone and power access. To make the working area clean all the technical area was placed at the back of the desk.

Overall, concept 1 focused more on modularity, hidden functionality and visual integration. The main strength of this concept was that it combined several features and functions into the desk surface such as cable management, storage, monitor support and control interaction. However, the concept had its limitations like complexity that came with lifting panels, monitor riser mechanisms integration into the desk itself, hidden storage increasing the thickness of the tabletop and manufacturing cost of the desk. The visual appearance of the desk in this concept was on the bulky side that might affect the performance of the height-adjustable frame.

4.7.3 Concept 2

Figure 7

Concept 2



The above figure 7 shows the concept 2 generation. The concept was based on simple and sleek design with the rear end having the main integration slot. The rear end of the desk has an accessories integrated slot in which monitor riser can be placed and removed and behind that is a casket box type cable management. The desk consists of keyboard and mouse tray that can be used to storage them away when not in use. The left and the right phase of the desk has small inside groves which can be used to attach different things such as multipocketed bags, headphones, cup holder and small storage bucket. The rear end section of the desk also has sockets and ports.

Overall, concept 2 had a lot of main requirements from the synthesis table as it addressed cable management, storage, accessibility, tabletop utility and visual organization. Compared with other concepts this was easier to develop further because of easier and clear functions that were easier to understand.

4.7.4 Concept 3

Figure 8

Concept 3



The figure 8 above shows the concept 3 generation and how the accessories can be placed on the tabletop. This concept takes inspiration and builds upon the most general and common solutions that exists in the market. The idea was to create a similar looking workstation with the visuals of the accessories trying to match the whole aesthetic of the table. The concept uses a continues rear spine where different elements can be attached. The rear spine acts as a shared space for the cable management, privacy screen, monitor support and power outlets.

The intention behind this concept was to reduce the visual clutter on the workspace at the same time keeping important tool in the proximity area for the user. The privacy screen gives the desk more enclosed and focused workspace. It also supports the idea of visual calm in the shared office environment where users want more privacy.

The concept also include accessories such as laptop stand, utility box, headphone stand and monitor riser. The accessories were designed to follow the same visual language as the desk,

using simple geometric shape and a dark finish. This directly connects to the aim of the study of creating a coherent workspace instead of separate attached products to a desk.

The sliding tray under the tabletop was created to hide the keyboard and mouse to free up the surface and get more space to support clean work area. The tray also acts as a storage space for other things like stationery items, sticky notes, documents and some measuring instruments.

The concept 3 consist of several design requirements that came from the synthesis table. It supports cable management, privacy screen, accessories, clean layout and flexibility to re-arrange things that can be placed or adjusted according to the user's needs.

However, the concept also showed limitations that needed further development. The rear end of the desk and the privacy screen made the overall look of the desk visually heavier if the dimensions were not carefully controlled. The attachment mechanism of the accessories also felt left off and needed to be modified so that it easy to use, stable and not complicated.

4.8 Concept Selection

The concept was selected after presenting them in front of the company stakeholders. A presentation was done in which the whole progress of the study was shown with the 3 concepts and at the end feedback was taken from the representatives from ROL Group and the company supervisor. The purpose of this was to make sure that the selected concept was aligned with the expectations and practical understanding of the company. Since, the project was done in collaboration with ROL Group, their feedback was relevant and important.

4.8.1 Rating and feedback

Table 4

Concept selection

Concept	Rating by P1	Rating by P2	Rating by P3	Main feedback	Decision
Concept 1	1	1	1.5	Too bulky and complex	Not selected
Concept 2	4	4	4.5	Strongest overall direction with good potential for cable management, utility, and integration	Selected
Concept 3	3	3.5	3.5	Some useful features, especially accessory and privacy screen	Features partly included

After concept presentation the concepts were asked to be rated on the scale of 1-5, where 5 being the highest and 1 showing the weakest direction. Table 4 above shows the result of the concept selection where P1, P2 and P3 are people from ROL Group. After discussions and feedback, the concepts were rated and one of the concepts was selected. Concept 2 was selected as the clear design direction with some features that could be added from concept 3

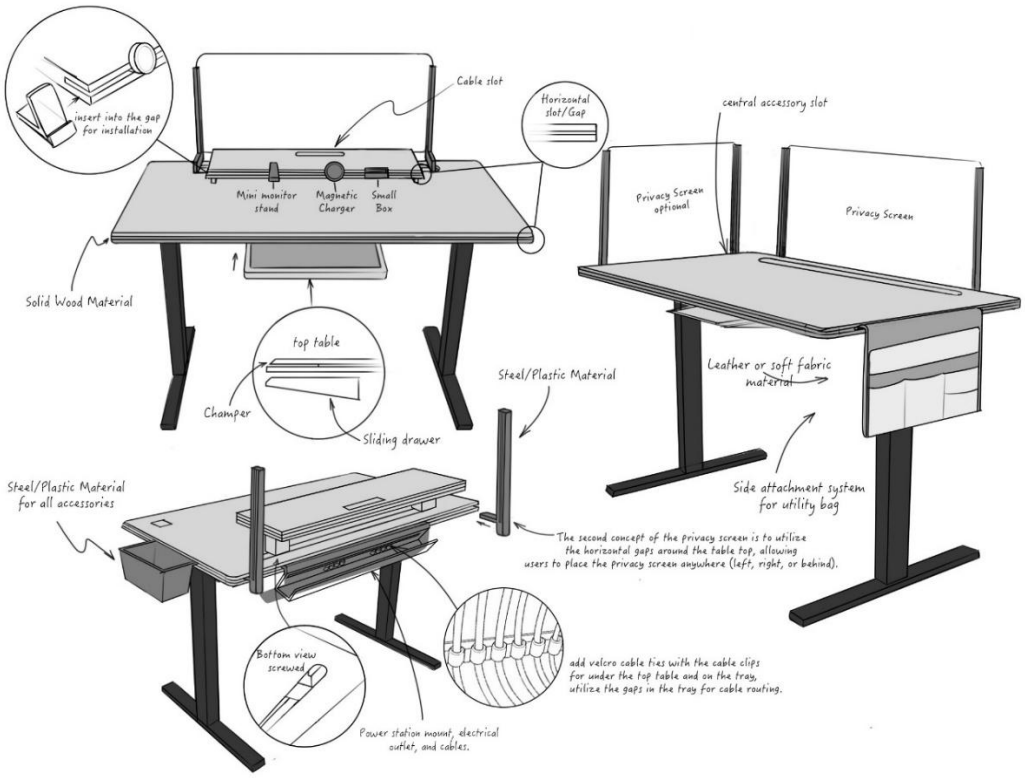
4.8.2 Integration of features from other concepts

While concept 2 was chosen as the final direction, there were some aspects from other concepts and refinements in the concept 2. This was done to strengthen the selected direction and make it more complete. The concept of rear cable management remained unchanged, but the concept of accessory management was refined in such a way that it became a part of the overall desk rather than separate element. The desk surface was considered too bulky for the concept 2 as well, so in the refined concept the thickness was also reduced. These changes helped to develop the concept in a more coherent visual language.

4.8.3 Refined Final concept

Figure 9

Final Concept



The main idea of the refined final concept is to use the tabletop as a system instead of a flat surface. In figure 9, the tabletop is divided into three layers and there is an intentional gap between the top surface and the bottom surface of the tabletop. The gap appears as if the desk is sliced in half, but it is attached in the middle so that the two surfaces are not floating in air but are connected through the middle surface with some thickness. With the intentional gap it becomes easier to connect accessories around the tabletop with just sliding them inside the gap wherever the user feels like. Example, headphones hanger accessories can be placed on the right side of the desk, utility basket can be placed on the left or the right side of the desk. The same sliding mechanism works for the cable management tray and privacy screen holder columns. The monitor riser can be connected to the accessories slot located at the rear end of the desk surface. The monitor riser also has the same horizontal cut in which other

accessories like iPad stand, magnetic charger and small storage box can be connected. There is a sliding tray underneath the desk surface to keep the keyboard and mouse when not in use. The sliding tray can also store other small items. Inside the cable management tray, the wires are organized with Velcro cable ties.

5 Final Design Concept

This chapter shows the final concept developed from the selected design direction. This concept is further developed on concept 2 that was selected during the concept evaluation process. Some features of other concepts were added to make the final solution more complete. The final concept focuses on creating the height-adjustable workstation where the tabletop, frame and accessories work together as one system.

The main design of the final concept is to fully utilise the tabletop instead of thinking it as a flat work surface. It is now designed to work and support various accessories, cable management and privacy screen. The concept aims to reduce visual clutter, improve cable management and at the same time support task related comfort and visually create a coherent workstation. The final concept was designed in 3D software SolidWorks.

5.1 Concept Overview

Figure 10

Final concept 3D model



The above figure 10 shows the final concept of the desk. The concept focuses on creating a cleaner and well-organized workplace using the desk accessories, cable management, storage and privacy solutions into one system. The idea is built on combining the height-adjustable frame with a redesigned tabletop system. Instead of a flat working surface, the tabletop is treated as a functional part of the workstation consisting of accessory gap, integration zone, cable management system, monitor riser, sliding tray, privacy screen and storage space. The aim is to reduce the need for random add-ons which takes unnecessary separate space on the desk and create a more well-organized desk which allows different elements to be attached and organized all around the tabletop. This gives the user an option to decide where and how close they want their important tools and devices to be while maintaining the work surface clean.

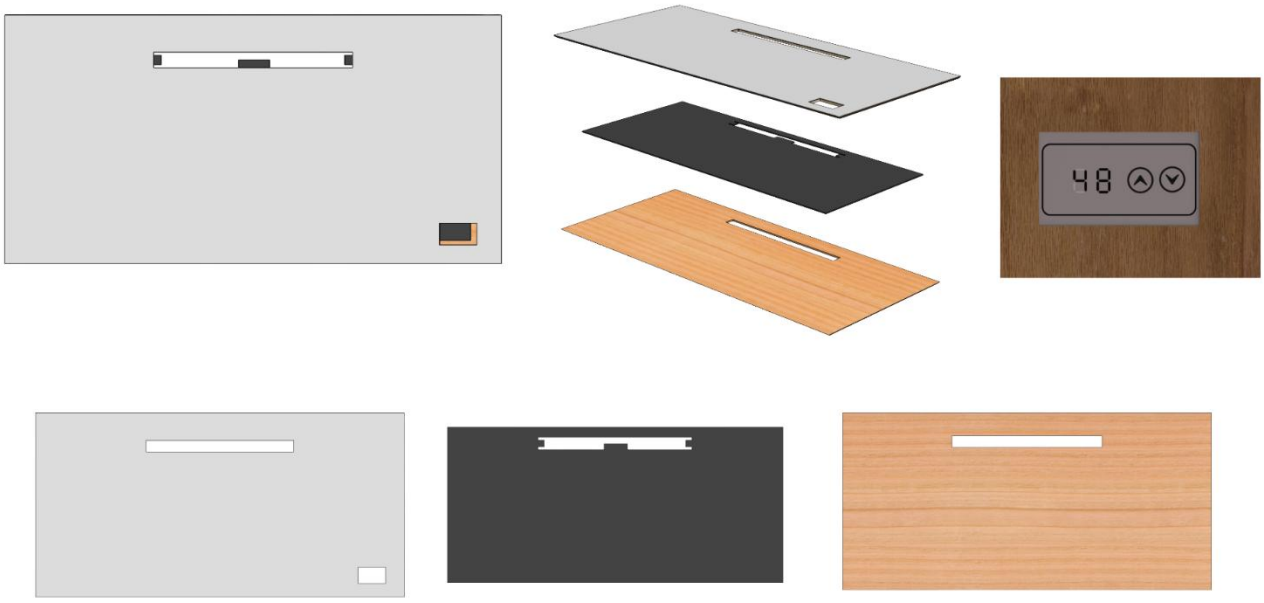
5.2 Design direction and visual language

The concept has a minimal and functional design approach. As the project is linked with the office furniture, the design is intended to be a professional workspace but not too technical or heavy as an office furniture. The goal was to create a design which will give calm, organized and a suitable modern work environment. Straight lines, soft edges and horizontal layers help give a clean visual language. The horizontal gap in the tabletop provides an intentional design detail instead of only a functional cut-out. This gives the product an identity of its own as well as supporting accessory attachment and cable organization. The accessories are designed to follow the same visual language of the desk. This was important as one of the problems found during the research was that many desk accessories feel like separate add-ons. In the final concept, everything is designed to feel like they belong to the same product. The overall design is kept minimalistic which prevents the desk from appearing cluttered due to overloaded features.

5.3 Tabel Top system

Figure 11

Tabletop parts and assembly in 3D model



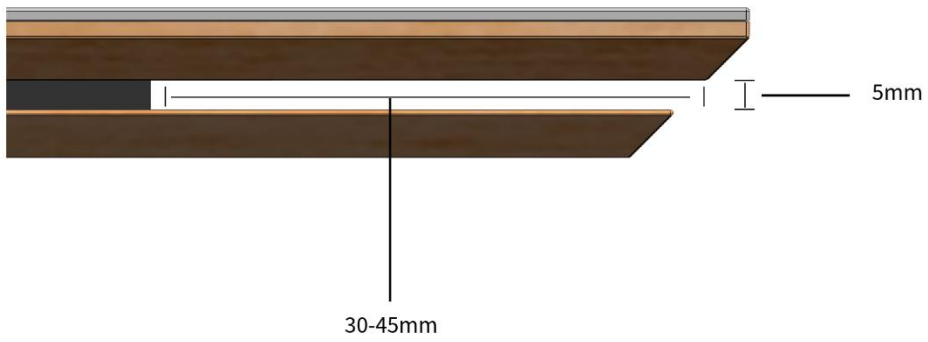
The tabletop is one of the main parts of the final concept as it is designed to support the desk work while connecting other elements to it at the same time. From above figure 11, the tabletop gives the option to have four phases to interact with right, left, front and rear to the user. The tabletop is made of 3 layers combined in a sandwich way to look like a single tabletop. It is divided into 3 layers where the top surface is of 9mm, the middle surface where the gap is of 5mm and the bottom surface is 9 mm. The overall dimension of the tabletop is 1600X800mm with 25mm of thickness. Even though the thickness is a little more than some of the standard thickness of tabletop on the market, the overall visual look of the desk is not that bulky. There is a chamfer going downwards which gives the desk a sleek look when viewing from a straight or top angle. The front part of the tabletop is for the daily desk work while the rear section is for cable management, monitor support, privacy and storage. There are fillets on the edges which makes the tabletop soft to improve comfort during use. There is a rectangular slot on the tabletop for the handset to be placed there instead of having it underneath the

tabletop. The handset is used to control the sit-stand function and the new position makes it easier to see and reach.

5.4 Horizontal accessory gap

Figure 12

Horizontal gap magnified view



The gap shown in the figure 12 above is an important feature of the final concept. It is all around the tabletop from left to right and from front to back. The gap is of 5mm in height and 35-40mm in horizontal depth. The gap allows the user to attach and position the accessories without any clamping or separate mounting system. It is up to the users will and they decide where they want to put the accessories according to their reachability and needs. This makes the desk more adjustable to different work as the desk is more adaptable to different users and the gap can be used for things such as storage box, headphones hanger, utility bag, laptop bag hanger, privacy screen, cable management tray and other accessories. This gives the user a flexibility to control the desk and makes the desk look clean and easy to use because the accessories are still connected to the desk and are not randomly placed on top of the desk. This supports the current study of creating an integrated workstation were

desk and accessories work together. The gap is also the part of visual identity. It gives a lighter and clean design detail that fulfils the functional purpose as well.

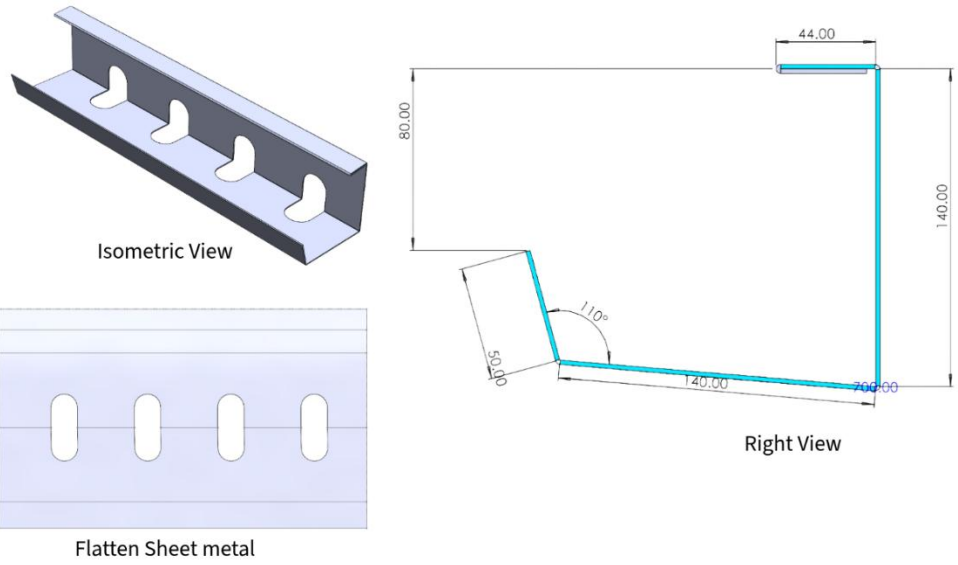
5.5 Rear integration zone

The rear zone is the main technical area of the desk which includes the cable slot, monitor riser, accessory slot and privacy screen placement. When all these things sit at the back the main working area of the tabletop feels cleaner. The rear section also helps with the layout because it holds elements that are major cause of clutter like cables from monitor, laptop, phone charger and power outlets. Instead of having those items around the surface they are gathered into one area. This section helps with the accessories too. For example, the monitor riser sits at the rear area, and the cables can pass through behind it. The user sees the desk as having the front working area and rear support area. This idea makes the desk easy to use and looks more organized visually.

5.6 Cable management

Figure 13

Cable management Tray in 3D



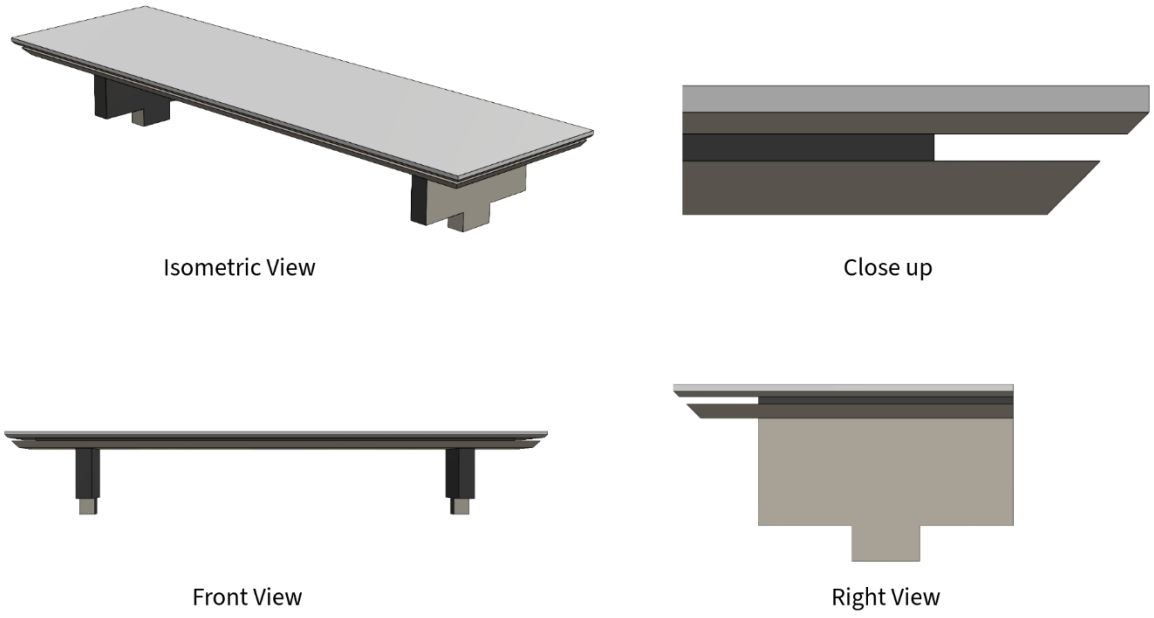
Note. Cable management Tray in different view

Research and observation showed that cables are often become visible and tangled in normal desk setups. For this reason, a cable management tray was introduced in the final concept that works through both the tabletop and underside of the desk. There is a horizontal accessory slot on the top of the desk which goes all the way through the tabletop allowing wires from monitor, laptops, chargers and other devices to pass through and sit inside the cable management tray. Inside the tray the cables can be managed using clips, cable ties or Velcro wraps. A power extension is held inside the tray which connects all the devices sockets and provides power. The aim is to reduce the visual clutter of cables and hide them. This solution supports both visual organization and practical use. It helps keep the desk surface cleaner and reduces the mess usually created by multiple devices and chargers. Figure 13 above shows the cable management tray in 3D model.

5.7 Monitor Riser

Figure 14

Monitor Riser



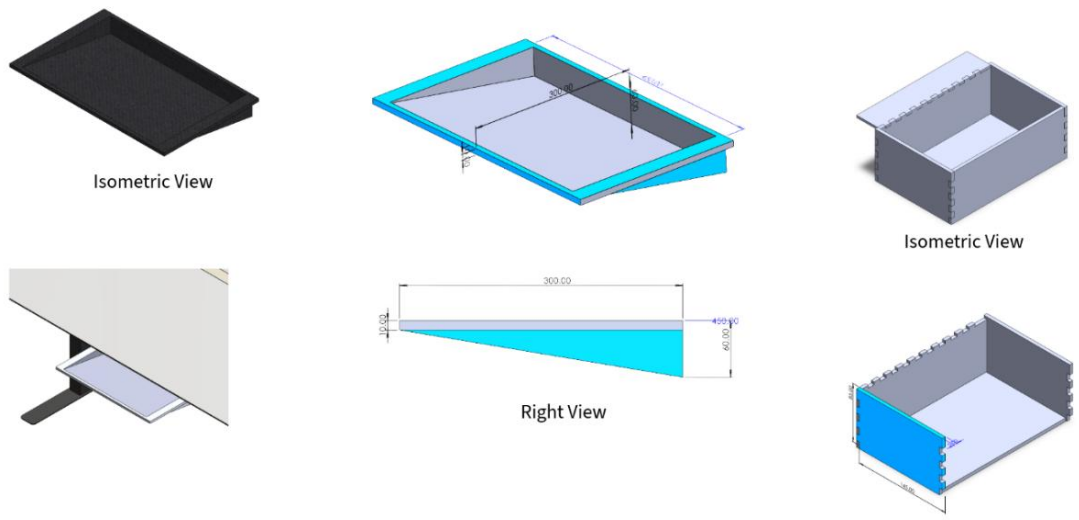
Note. Monitor Riser with various view

The monitor riser shown in the figure 14 above is one of the accessories and is designed to support screen area and create a more organized workstation. It follows the same visual design language and detail of the tabletop and has the similar gap to insert and different accessories like a wireless phone charger or iPad holder. There monitor riser sits on the accessory slot gap which is at the rear section of the desk. It also creates space below it which can be used for small items or cable routing. The aim is for the riser to feel like a part of the desk rather than a separate object placed on top. This feature also supports the requirement for better accessory integration. Instead of using an external monitor stand that may not match the desk, the riser becomes part of the overall workstation system.

5.8 Sliding keyboard, mouse and storage tray

Figure 15

Sliding tray and Storage box in 3D model



Note. Different view of sliding tray and storage box

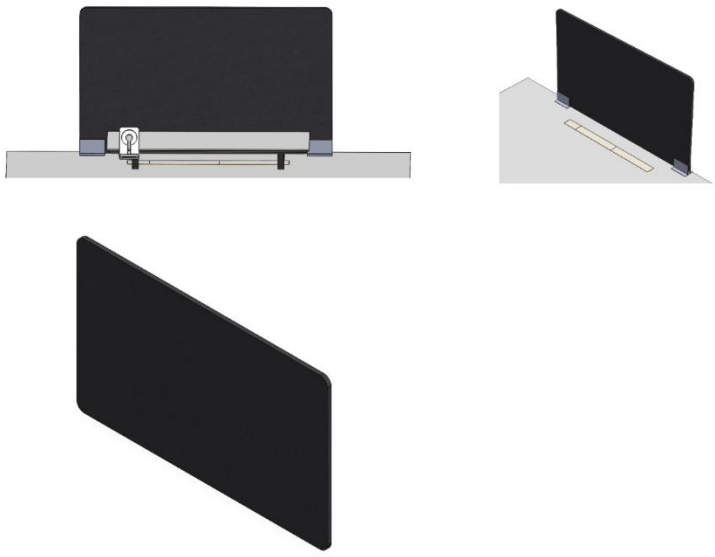
Sliding tray is placed under the front section of the tabletop. They tray is used for storage of keyboard and mouse when they are not in use or depending on the user's needs. This helps

the tabletop maintain the free for other work tasks such as writing, sketching, or using laptop. An option for storage is also included in the final concept which gives the user a small storage feature such as a utility basked where everyday items like chargers, spectacles, notebooks, pen paper, cables and personal objects can be kept. The research shows users use items kept on the desk surface for quick access, but when too many small items are kept on the tabletop, the desk starts to look cluttered. The storage features are meant to give the final concept a defined place for these objects. The storage is not designed to replace the large office storage instead support everyday light organization. The above figure 15 shows different views of the sliding tray and storage box in CAD model.

5.9 Privacy screen integration

Figure 16

Privacy screen in 3D model



The final concept includes an optional privacy screen which can be attached to the desk same way the cable management tray is being attached with the slide system inside the gap space at the rear end of the tabletop. The privacy screen is for the focus mode inside the shared office space environment, and it helps to reduce visual distraction and gives the user a

personal work area. The screen can be chosen to be used or not depending up on the work environment. It also defines the rear boundary of the desk. Figure 16 above shows the privacy screen in 3D model software.

5.10 Material and finish direction

The final concept combines warm materials such as tabletop is made of Birch plywood material to give a warmer natural surface to the user. The accessories are made from steel or plastic depending on strength, cost, weight and manufacturing requirements. The finish direction should combine neutral colours, wood tones and darker technical details that can help the desk look professional while still feeling warm.

5.11 Relation to design requirements and Summary of Final concept

The final design concept connects to the main design requirements developed from the research and synthesis phase. To reduce the visible cable clutter and improve the access a cable management tray is used which helps organize cables in a more controlled way. For the accessories integration, the horizontal gap allows different accessories to be attached to the desk and simple slide mechanism with the help of friction to allow the user to place the accessory according to their needs while keeping connected to the system. The monitor riser, sliding tray and accessory placement responds to the requirement of better task related ergonomic support and better workstation layout by helping the everyday tools, screen, keyboard and mouse organization in a cleaner way. The side utility storage gives the user an option to keep personal items and small work tools without taking space on main tabletop. The privacy screen can be used when privacy is needed, and it responds to the need for a more focused workspace, and it can be removed when it is not required. The final concept gives a strong base for the next stage of the study, where the design is tested through working prototype evaluation. The evaluation stage will focus on how well the integrated features of the working prototype perform in practice with respect to reach and task related comfort.

Figure 16.1

Poster shot of the workstation



6. Prototyping and Evaluation

6.1 Purpose of the prototype

A physical working prototype was created for evaluation of functionality of the final concept. The primary aim was not to fully mimic the production quality rather understand how the concept physically works as far as its mechanics are concerned. It was important as some of the components like monitor riser, accessories and storage needed to be understood practically and not only through CAD models. The prototype helped in the testing stage to evaluate the basic interaction between the user and the desk system such as accessibility, reachability, placement of accessories and overall usability of the system.

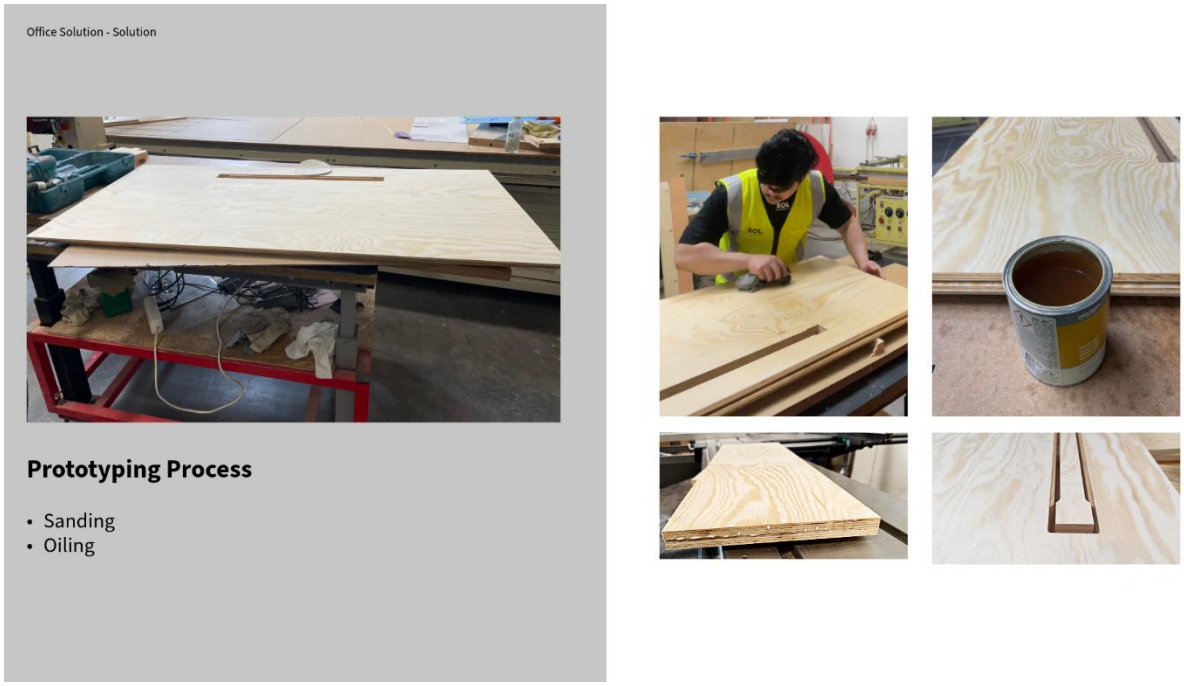
6.2 Prototype Development

The prototype was built as a functional and full-size model with height-adjustable desk frame as the foundation, while the tabletop and other integrated component were designed and fabricated independently. Mainly the tabletop was made using plywood allowing construction, configuration of slots and other surface details during the building process. The prototype included main features of the final concept such as tabletop system, monitor riser, cable management tray, accessories and storage. However, some details have been simplified compared to the final CAD model while keeping the key interaction points intact in order to perform the tests. The purpose of the prototype was to test concept functionality and clarity to understand whether the design direction was practical and clear in real use. It also helped in discovering minor details which were hard to imagine only from sketches, CAD and rendering.

6.3 Prototype Building Process

Figure 17

Prototyping process

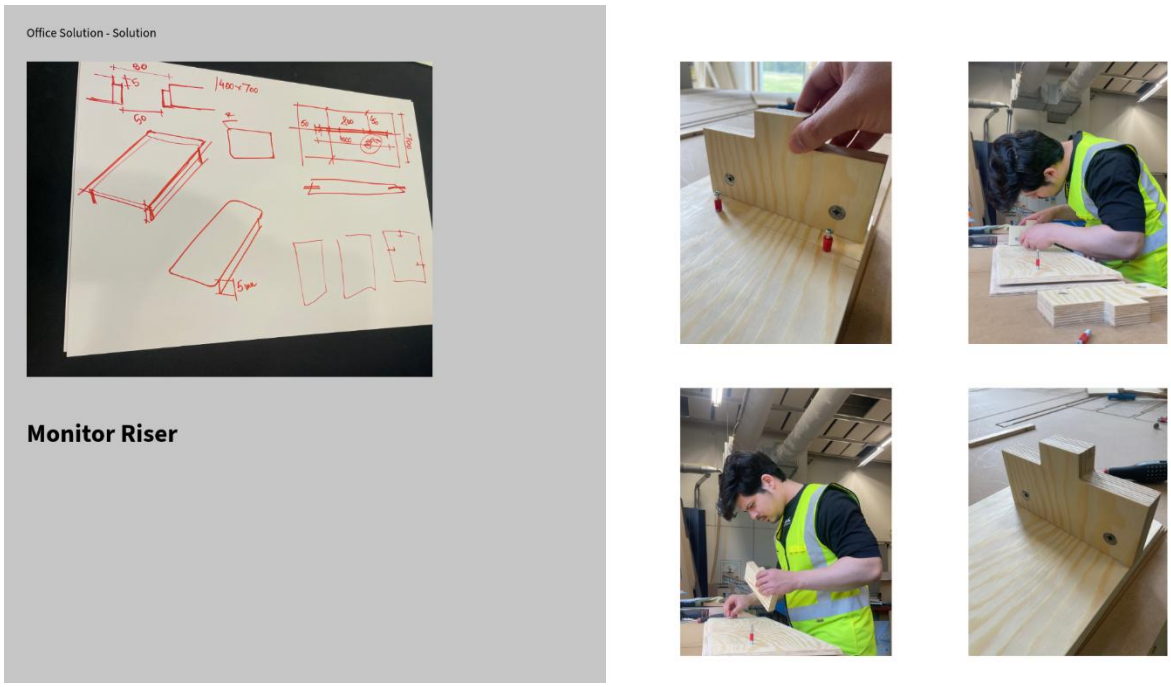


Note. Prototype building process showing the plywood tabletop, sanding, oiling, and slot preparation

Figure 17 above show the process started with preparing the tabletop by cutting two Birch plywood board and glueing them together and then cutting the required tabletop size and creating the main slots based on the final concept. The horizontal slot and the cable slot or slot for monitor riser was important because they represented the main integration system of the design. The plywood surface was sanded to improve the quality and remove rough edges and was then surface treated with oiling process to improve its look and make the prototype look closer to the finish product.

Figure 18

Monitor Riser assembly

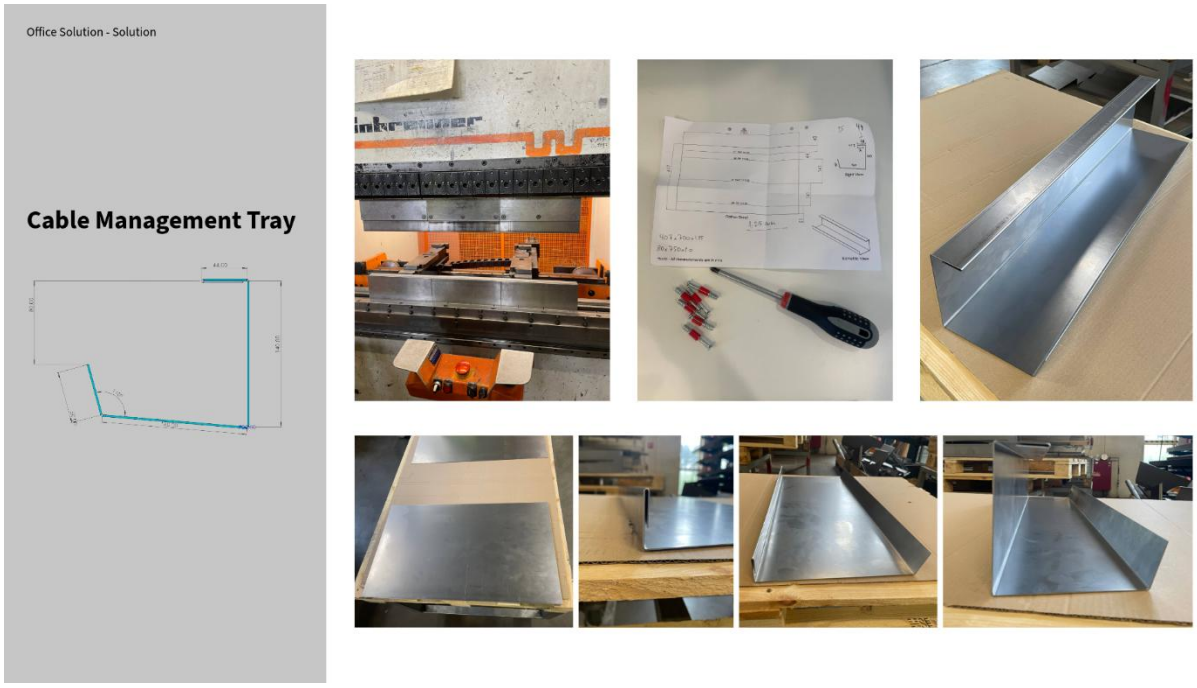


Note. Development of the monitor riser, including sketching, measuring, fabrication, and placement on the tabletop

The initial measurement along with the monitor riser slot on the tabletop helped in identifying the size, position and height of the monitor riser. After that, monitor riser was made from the same Birch plywood as the tabletop and tested on the table to check its position and stability. The main purpose of the riser was to support the monitor and create a more organized working zone behind the table. Figure 18 above shows the assembly of monitor riser prototype.

Figure 19

Cable management Tray Fabrication

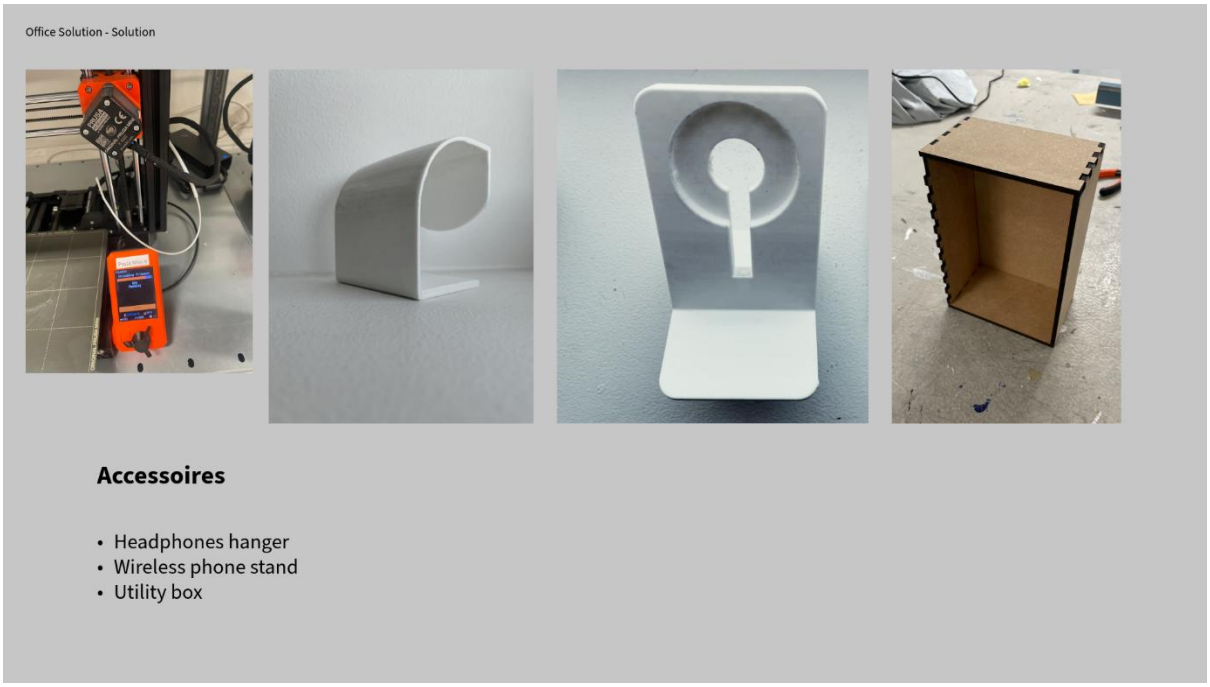


The cable management tray is made from sheet metal. Material steel DC01 of thickness 1.25mm is used for the tray. The tray is supposed to be attached below the rear part of the tabletop, gathering all the cables and power components in one place. The above figure 19 shows several steps of how the tray was bend starting from a flat surface and ending with several edge bend surfaces. This part was also crucial in prototype as cable management became one of the main discoveries during the research stage.

Additional accessories such as headphone hanger, wireless charging phone stand and utility box for storage were also created to determine whether the accessories would integrate into the system well or not. Different process was used to create these accessories, which included 3D printing, laser cutting glueing, sanding and spray painting some of the accessories. These accessories helped in testing stage showing how smaller elements could work with the desk system and support everyday organization. Figure 20 below shows the accessories.

Figure 20

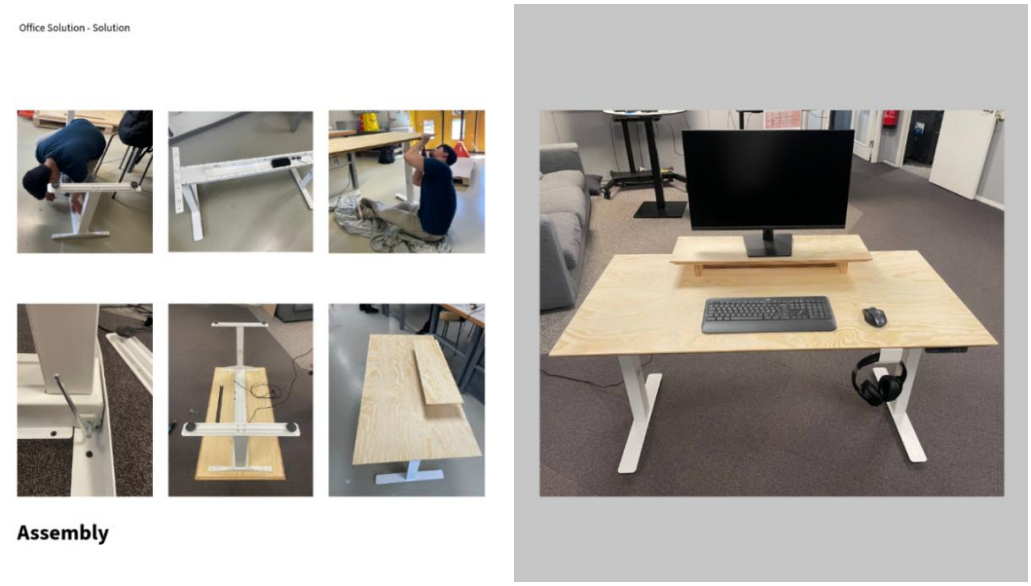
Prototype of accessories



Note. Prototype accessories including headphone hanger, wireless phone stand, and utility box.

Figure 21

Assembly of working Prototype



The figure 21 above shows final assembly was done by assembling the height-adjustable frame legs or columns, attaching the tabletop on top of it and adding the monitor riser, cable management tray and other accessories.

6.4 Prototype Components and Final assembly

The final working prototype consists of the following components

- Birch Plywood with horizontal slot and cable slot for riser
- Height-adjustable desk frame from ROL Group
- Monitor riser made with Birch Plywood
- Sheet metal cable management tray
- Headphones hanger
- Wireless phone stand
- Utility Box

In next stage features and components from the final working prototype were tested to check whether the desk looked organized, logic behind the placement of accessories was easy to understand by the user or not and if the desk followed a low ergonomic risk layout design.

6.5 Evaluation Setup

After the assembly of working prototype, a usability and ergonomic analysis was carried out. The aim of this test was to understand how users interact with the prototype and how well the main design features supported task-based comfort, desk organization and everyday usability. The evaluation was done in three parts. First, a task-based usability test was carried out by participants using the prototype. Second, a ROSA assessment was done for certain aspects of ergonomics for the desk setup, focusing on monitor, keyboard and mouse placement and usage. Third, a SUS questionnaire to collect feedback from the users who interacted with the prototype. The evaluation focused on all the main aspects of the concept, including the monitor riser, the accessory system and keyboard and mouse placement.

6.6 Task-Based Usability Testing

A task-based usability test was done to understand how users interact with the working prototype. The purpose of this test was not only to ask the participants what they thought about the working desk prototype but to also observe how they behaved with it. It was important as some usability are easier to notice when users perform tasks rather than only answering questions. The test was done inside the company office of ROL Group at Jönköping, Sweden. Four participants were asked to perform realistic tasks using the working prototype while sitting down in a chair. The task was to interact with the desk use keyboard and mouse, reaching for accessories like headphones hanger and wireless phone charger. After their interaction informal questions were asked to understand what the users found useful or confusing. The figure 22 below shows the pictures of the users performing the test.

Figure 22

Task-based usability testing photos



Note. Task-based usability testing with participants interacting with the prototype and evaluating the desk layout, accessories, and workstation setup

The test focused on the following points:

- whether users understand the desk features
- whether the accessory system felt useful
- whether the user felt confused with the accessory system
- whether the user twist their body when using the accessory system

- whether the user lifts their shoulders or arms too much when using the accessory system
- would the users like to use this desk in real office
- would the users like to change something
- a simple user rating from scale of 1-5

From the observation and the answers received after asking some informal questions the users were able to understand the main desk idea without any confusion. The accessory sliding system was seen as a useful way to organize items wherever they wanted it to be, and it reduce clutter on the tabletop at the same time. Participant could imagine using the desk being used in real office environment, especially when the desk or the working prototype combines the sit-stand function with more organized storage and accessory placement. Some improvements were suggested related to the size of the monitor riser and certain features. The feedback was important for the future development of the product. Users also quoted that the surface finish of the tabletop gave them a tactile feeling and a sense of calm. A simple user rating was received after the task-based usability test and an average rating of 4.6 out of 5 was given. Since, the participants were limited, this result should not be understood as statistical representation but an early understanding of the design and feedback.

6.7 ROSA Ergonomic Assessment

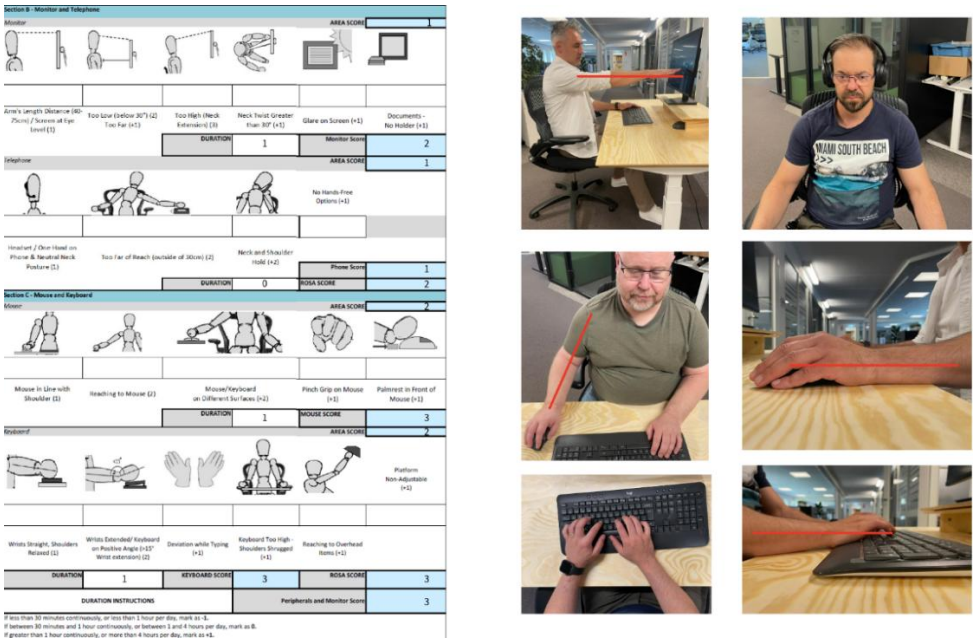
Another method used for usability testing was the use of Rapid Office Strain Assessment (ROSA), where risks due to office workstations configurations like monitor, chair, telephone, keyboard and mouse setup are considered. For this study ROSA assessment mainly focused on monitor and other peripherals such as mouse and keyboard. The focus was on these components as they were directly related to the working prototype that could be evaluated to check whether the workstation layout created major ergonomic concern or whether some parts needed improvement. Photos were used during the assessment to confirm the position

and posture of the users. The monitor was examined with respect to the viewing distance and height while the keyboard and mouse were tested in terms of the arm position and wrist posture.

The figure 23 below shows a score of 3 which means low level of ergonomic risk. It also shows the images of participants and how the monitor in relation to viewing distance and height was assessed. The mouse and keyboard were tested in terms of arm position, wrist posture and whether the user had to reach for them or not. The duration score of 1 was added to each participant as expected daily use of the desk is more than 4 hours daily.

Figure 23

ROSA Assessment Images



Note. ROSA assessment of monitor, keyboard, and mouse position during prototype testing

In total of 4 participants did the Rapid Office Strain Assessment (ROSA) the test was conducted in ROL Group company in Jönköping, Sweden. The result shows a basic ergonomic check rather than a complete validation of the working prototype. The test was

done partially as the design of the final product did not include the chair as a part. The chair was only used to allow the users to sit and interact with the working prototype in a normal office working posture. The assessment was helpful as it identified the layout of the workstation prototype supported the accepted working posture. The table 5 below shows the score.

Table 5

ROSA score

Participant	Monitor and Peripherals ROSA score	Main observation
P1	3	Mouse reach and wrist extension
P2	3	Mouse reach and wrist extension
P3	3	Similar observation as above
P4	2	Minor adjustment

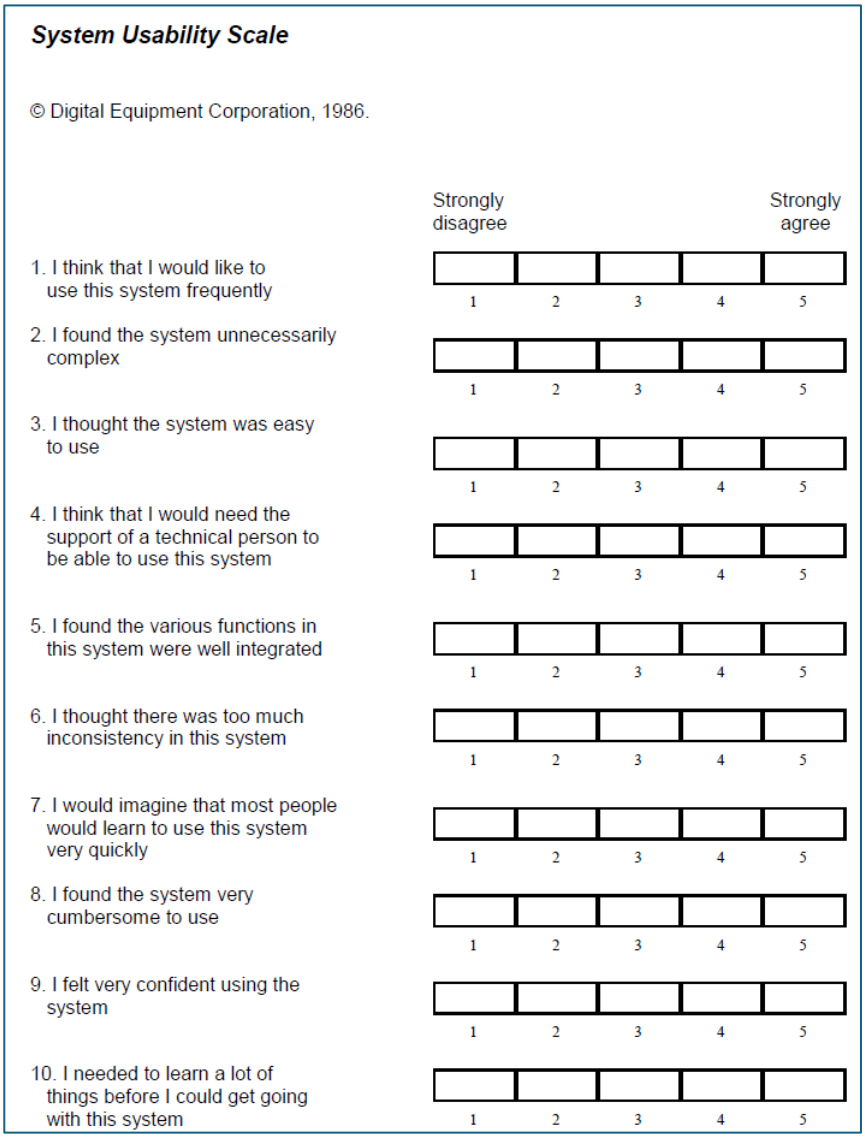
6.8 System Usability Scale (SUS) Evaluation

SUS was conducted at the premises of Jönköping University, Sweden. The aim was to get a usability score quickly and easily from more participants after they interacted with the prototype. Participants were invited to test and view the prototype and then answer the SUS questionnaire through QR code. The QR code was linked to a google form allowing feedback to be collected in a quick and organized way. The result from the SUS helped in supporting the task-based usability finding. The form included validated questions that focused on the overall usability of the desk prototype, usability of the desk features and how confident users feel while using the desk. The figure 24 below shows the ten questions and the scale of 1-5 in

which 1 is strongly disagree, 2 is disagree, 3 is neutral, 4 is agree and 5 is strongly agree. The form was filled by 13 people with their consent who first interacted with the working prototype of the desk and then immediately filled up the form without thinking about the questions for a long time. The table shows the rating of the users. To calculate the SUS score minus 1 was deducted from score of the odd or positive question such as Q1, Q3, Q5, Q7 and Q9. For negative and even questions such as Q2, Q4, Q6, Q8 and Q10 the score was 5 – answer. Then total adjusted score multiplied by 2.5 equals SUS score out of 100.

Figure 24

System Usability Scale



Note. System Usability Scale Validated Questions

Table 6

SUS score calculation results

Participant	Adjusted SUS Total/40	SUS Score/100
P1	40	100
P2	37	92.5
P3	34	85
P4	36	90
P5	37	92.5
P6	36	90
P7	40	100
P8	38	95
P9	37	92.5
P10	40	100
P11	39	97.5
P12	40	100
P13	40	100
Average	38	95

From the above table 6, the result shows that users found the workstation easy to understand, simple to use and well-integrated but it should not be understood as statistical generalization because of the small scale of participants.

6.9 Summary of Evaluation Findings

The evaluation showed that the prototype is understood by the users and the overall direction of the concept remains clear. The results from the task-based usability test showed that participants could image the working prototype of the desk in a real office environment

and understood the system as a functional means to organize the workstation. The assessment from ROSA gave a score of 2 for the monitor and peripheral, which indicates a low risk of ergonomic concern with the workstation elements while at the same time it showed there can be further refinements in the design for long-term use and adjustability. The SUS score of 95 shows a good usability, which also suggests that user found the desk system easy to understand, well integrated and simple to use.

Figure 25

Prototype in real office scenario



7 Discussion

In this chapter, the outcomes of the current study will be discussed in relation to the aim, research questions, design requirements, and evaluation outcomes. The aim of the thesis was the exploration of how a height-adjustable desk can be developed as an integrated workstation system that supports task-based comfort while reducing the clutter and improve usability for office workers. The project started with the height-adjustable desk that already solves one problem which is switching between sitting and standing allowing the users to change their position. However, the research shows that the sit-stand function alone does not solve the full workstation challenges that are associated with the workplace. Office workers still use separate accessories for cable management, storage items, monitors, chargers and personal objects during their daily routine. Adding these components helps the user but at the same time it also takes up space on the tabletop. Since, each component is added separately the workspace becomes cluttered and this problem was the main reason for developing a more integrated desk system for this study.

7.1 Discussion of the research questions

The first research question. What problems do office workers have with accessories, cables, storage and comfort in their current workstation setups?

The findings identified from the literature review, market analysis, interviews and observations showed that office workers experience several problems. Cable management was one of the most prominent concerns. Cables were often visible behind the monitor, under the table surface or even on the floor. This affected the visual quality and the ease of management of the workstation. The question of storage and organization was another important problem. Users often kept their personal items on the surface of the desk such as pen, phones, headphones, water bottles because of accessibility creating visual clutter and reducing the usable works area of the desk. Another problem was accessories that were

bought separately as the solution to individual problems. Theses accessories were often added later separately that may solve users' problems, but they do not always work together as one system. This gives visual impression of inconsistency within the workstation. Such items include monitor stand, headphones hanger, cable management tray, storage box. Task related comfort was also connected with working at desk. Even when desk has the feature of sit and stand the placement of the monitor, keyboard, mouse and other accessories can still affect the reach, posture and task related comfort. This means that ergonomic should not only support desk height but also on how the whole workstation is arranged.

The second research question. How can integrated height- adjustable desk system be designed to reduce visual and cable clutter while improving task related user comfort?

This was answered by the final concept by developing the desk from the start as a complete workstation system rather than having only one sit-stand function. The design of the tabletop with an intentional gap and rear integration zone that hold the monitor riser and could have a flush look when not in use. The horizontal intentional gap severs the purpose of multiple things such as an integration accessory gap, rear cable management solution, privacy screen and storage feature. This allows the user to adjust accessories according to their needs and it gave them the flexibility without making the desk feel like a collection of separate add-ons. The cable management tray integrated with the same logic reduces the visible cables by guiding them through the tabletop and under the desk. The storage features like the utility box gives daily items their defined place. The design also supports a cleaner workstation layout with the help of monitor riser and sliding tray. Overall, the final concept does not only add more features but integrates the features with a common visual and functional language.

7.2 Reflection on the final concept

The final concept was based on the refinement of selected concept 2, which received strong feedback during the evaluation done with ROL company. The refinement was done by adding

useful features from the other concepts. This helped in creating a better solution which connects well with the synthesis table and design requirements. One merit of the final concept is that it creates a link between the desk and the accessories instead of placing accessories on the tabletop surface, the design provides a specific area where the accessories can be attached and placed depending upon the users reach or preferred area of placement. This supports the idea of integration as the accessories are not permanently built into the desk rather, they belong and connect to the desk. Another merit is cable management which was observed as one of the most common visual problems during the observation method. This was solved with adding a cable management tray without any clamping method. The cable management is not an afterthought with the final design as there is a guided route for the cables at the rear section of the tabletop. The storage solution is not designed to replace the large office storage, but it helps in organizing small personal and work-related items of the users for quick access. However, the final concept needs more detail testing to check for long term durability, stability and ease of use. The final concept is not a production ready product, but it shows the design direction of a complete office desk solution.

7.3 Evaluation of prototype and testing

The working prototype helped in the testing phase because building the prototype made it possible to understand the scale, reach, accessory placement, position of the monitor riser, sliding function and cable organization. Some things that were difficult to figure out in CAD became clearer once the prototype was built. The task-based usability testing showed how users interacted with the prototype and understand the features. An average rating of 4.6 out 5 gave positive feedback about the concept during the early evaluation. The partial Rapid Office Strain assessment score of 3 suggests that the workstation did not create a high-risk ergonomic concern in relation to monitor and its peripheral placement and reach. The main ergonomics to improve were related to mouse reach and slight wrist extension when using

keyboard. The testing showed difference between user perceived comfort and ergonomic scoring. Some users reported that certain position felt comfortable, while ROSA assessment showed that position could be improved. This shows why both are valuable in the evaluation. The System Usability Scale result collected at later stage provided useful information as users were able to easily understand and use the concept prototype. However, the result should be interpreted carefully and not treated as a full validation study.

7.4 Ethics

During the research and prototype evaluation ethical considerations were considered. Participants were informed prior about the purpose of the study, and their consent was taken at first, this was done especially during the user research and prototype evaluation. The workstation photos were only used inside the report and presentation within the thesis. Any personal detail such as name of the participants were not revealed anywhere including the report to maintain and protect the privacy of the participants.

7.5 Limitations

There are limitations in this study. The limited number of participants in the interviews, observations, and testing because of this the finding cannot be generalised to represent all office workers. Therefore, the result is a design insight rather than a statistically generalization conclusion. The working prototype does not represent the final production ready product. The prototype was useful in the testing stages, but it does not represent the manufacturing quality, final materials, durability, or any long-term performance. The limited ergonomic evaluation such as partial ROSA assessment. A full assessment would require a chair with more repeated use over time. The study mainly focuses on problems faced by office workers in connection to their workstation. Other things such as mass production, cost analysis, sustainability assessment, mechanical durability and safety were not evaluated as these were not inside the scope of the study.

7.6 Future development

The future development should focus on refining the prototype and making it production ready design. The stability of the accessory gap should be tested with proper tolerance and ease of attachment. The cable management system should be more accessible and should be developed in more detailed way that it could support different device setups. The sliding tray for the keyboard and mouse should be built in next prototyping and should be tested. A full ergonomic assessment in future can be done with the final version of the desk. The design could also be refined in terms of manufacturing, material selection, sustainability and modular repair. These areas would be an important step in the future development of the product.

8. Conclusion

The thesis shows how a height-adjustable desk integrating the frame, tabletop and accessories can be designed as a complete solution for office workers. The study focuses on reducing the visual and cable clutter while improving organization, and task-related usability comfort. The current study shows that workstation problem was not only limited to sit-stand function. Office workers experience other issues even if the sit-stand function was available. These issues were poor cable management, visible cables, lack of storage, third party accessories taking space on the tabletop, and inconsistent workstation setups. The findings from the research showed that the desk should be a part of a larger work system. The final concept was designed to solve these problems that can integrate the frame, tabletop and the accessories together to make it look like a coherent system. Together features like integrated tabletop system, horizontal accessory gap, rear cable management zone, monitor riser, sliding accessory system and privacy screen create a workstation where everything is more connected to the desk instead of being added separately. The working prototype and evaluation showed that the concept design direction was easy to use and easy to understand. The idea of more organized workstation received positive feedback from the users. The partial Rapid Office Strain Assessment (ROSA) showed that both the monitor and its peripherals were at low risk of ergonomic for users during the prototype test. Although some improvements are still needed. The major contribution of this study is the design direction for the workstation which takes the desk beyond the sit-stand feature. The study shows how cable management, accessory integration, storage, privacy and visual coherence can be achieved together when designing a complete office desk. The outcome of the study is the design concept of a workstation product with system integration, which can be used in real office environment. This study does not include fully production ready desk rather it provides a base for further development. Therefore, additional research is needed in future including

long term testing, detailed ergonomic validation, material, manufacturing improvements and refinement of both accessories and cable management systems.

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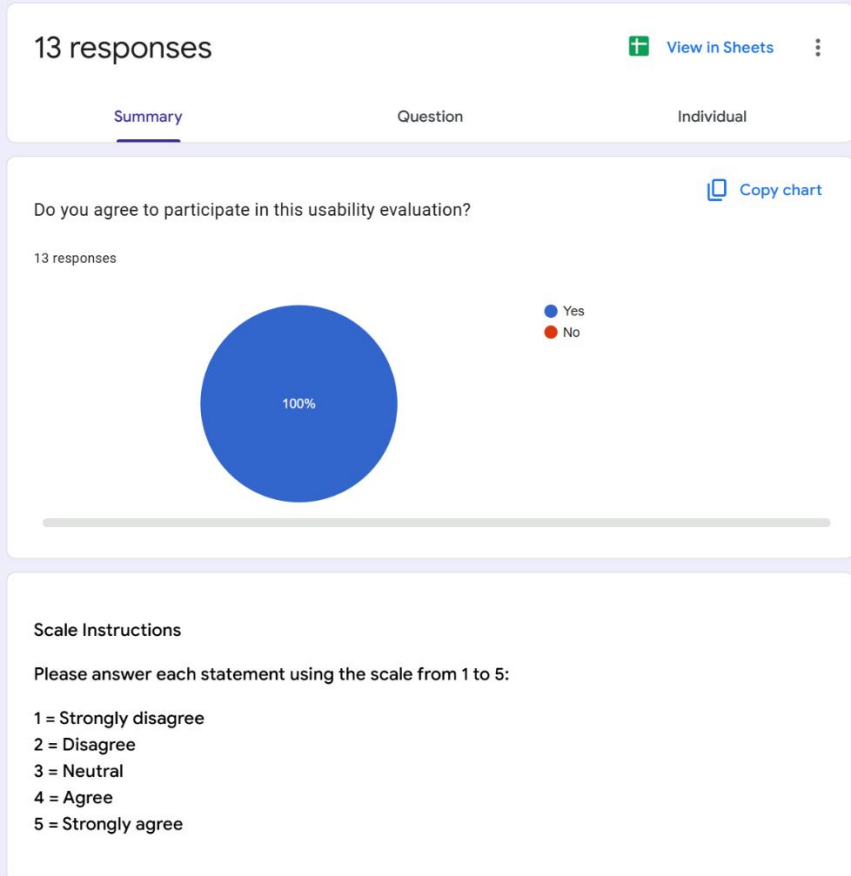
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Appendix

Systems Usability Scale form consent response



Informal Questions asked in Task-Based Usability Testing

Testing user 2

1) User understood accessories integration – yes
2) Was user confused – No
3) Did user bend forward when reaching the accessory – no
4) Did user twist body – yes, twisted the chair
5) Users lift the shoulders/arms too much – a little bit of arm
6) User comments – nice solution, maybe increase length of monitor riser to have two monitors
7) Would user use this desk in real office – Yes
8) What would user like to change – lid on the accessory slot to provide extra space with the riser
9) User rating – 4.8 always, comment of room for improvement

Extra Observation photos



Contextual Renders



